

**Institute of Engineering & Technology,
Bundelkhand University, Jhansi
BACHELOR OF TECHNOLOGY
IN
Computer Science & Engineering**



**Study and Evaluation Scheme
[Effective from the session 2021-22]**



Syllabus Of
B. Tech.
Computer Science & Engineering

Year 2nd, Semester-III

S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme				Subject Total	Credit
						Sessional		Examination			
		Theory	L	T	P	CA	TA	Total			
1.	2404 CS-301	Computer Organization	3	0	0	30	20	50	100	150	3
2.	2402 CS-302	Data Structures Using C	3	1	0	30	20	50	100	150	4
3.	2403 CS-303	Computer Based Numerical & Statistical Techniques	3	1	0	30	20	50	100	150	4
4.	2405 CS-304	Digital Logic Design	3	0	0	30	20	50	100	150	3
5.	2401 CS-305	Discrete Structure	3	0	0	30	20	50	100	150	3
		Practical/Training/Project									
6.	20406 CS-351	Computer Organization Lab	0	0	2	-	20	20	30	50	1
7.	20407 CS-352	Data Structure Lab	0	0	2	-	25	25	50	75	1
8.	20408 CS-353	Numerical Tech. Lab	0	0	2	-	25	25	50	75	1
9.	20409 CS-354	Digital Logic Design Lab	0	0	2	-	20	20	30	50	1
10.	20410 GP-301	General Proficiency	-	-	-	-	-	50		50	
		Total	15	2	8					1050	21

Year 2nd, Semester-IV

S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme				Subject Total	Credit
						Sessional			Examination		
		Theory	L	T	P	CA	TA	Total			
1.	2406 CS-401	Design & Analysis of Algorithms	3	1	0	30	20	50	100	150	4
2.	2407 CS-402	Database Management System	3	1	0	30	20	50	100	150	4
3.	2408 CS-403	Object Oriented System	3	0	0	30	20	50	100	150	3
4.	2409 CS-404	Software Engineering	3	0	0	30	20	50	100	150	3
5.	2410 CS-405	Theory of Automata and Formal Language	3	0	0	30	20	50	100	150	3
		Practical/Training/Project									
6.	20411 CS-451	Design & Analysis of Algorithms Lab	0	0	2	-	25	25	50	75	1
7.	20412 CS-452	Database Management System Lab	0	0	2	-	25	25	50	75	1
8.	20413 CS-453	Java Lab	0	0	2	-	20	20	30	50	1
9.	20414 CS-454	Software Engineering Lab	0	0	2	-	20	20	30	50	1
10.	20415 GP-401	General Proficiency	-	-	-	-	-	50		50	
		Total	15	2	8					1050	21

Year 3rd, Semester-V

S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme			Subject Total	Credit
						Sessional		Examination		
		Theory	L	T	P	CA	TA	Total		
1.	3401 CS-501	Computer Graphics	3	1	0	30	20	50	100	4
2.	3402 CS-502	Compiler Design	3	1	0	30	20	50	100	4
3.	3404 CS-503	Principle of Programming Languages	3	0	0	30	20	50	100	3
4.	3403 CS-504	Elective-I	3	0	0	30	20	50	100	3
5.	3405 HU-501	Industrial Economics & Principles of Management	3	0	0	30	20	50	100	3
		Practical/Training/Project								
6.	3406 CS-551	Computer Graphics Lab	0	0	2	-	25	25	50	1
7.	3407 CS-552	Compiler Design Lab	0	0	2	-	25	25	50	1
8.	3408 CS-553	Web-Tech Lab	0	0	4	-	30	30	70	2
9.	3410 GP-501	General Proficiency	-	-	-	-	-	50	50	
		Total	15	2	8				1050	21

* Chosen anyone from the followings:
(Web Technology, Data Compression, Embedded System, Semantic Web and Web)

Year 3rd, Semester-VI

Year III, Semester VI											
S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme			Subject Total	Credit	
						Sessional		Examination			
		Theory	L	T	P	C A	T A	Total			
1.	3406 CS-601	Operating System	3	1	0	30	20	50	100	150	4
2.	3407 CS-602	Computer Network	3	1	0	30	20	50	100	150	4
3.	3408 CS-603	Artificial Intelligence	3	0	0	30	20	50	100	150	3
4.	3409 CS-604	Elective –II*	3	0	0	30	20	50	100	150	3
5.	3410 CS-605	Organizational Behavior	3	0	0	30	20	50	100	150	3
		Practical/Training/Project									
6.	30411 CS-651	Operating System Lab	0	0	4	-	30	30	70	100	2
7.	30412 CS-652	Computer Network Lab	0	0	2	-	25	25	50	75	1
8.	30413 CS-653	Artificial Intelligence Lab	0	0	2	-	25	25	50	75	1
9.	30415 GP-601	General Proficiency	-	-	-	-	-	50		50	
		Total	15	2	8					1050	21

***Elective –II (Graph Theory / Operations Research / Quality Management / Product Development)**

Year 4th, Semester-VII

S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme				Subject Total	Credit
						Sessional			Examination		
		Theory	L	T	P	CA	TA	Total			
1.	4402 CS-701	Advance Computer Architecture	3	0	0	30	20	50	100	150	3
2.	4403 CS-702	Elective-III*	3	0	0	30	20	50	100	150	3
3.	4404 CS-703	Elective-IV*	3	0	0	30	20	50	100	150	3
4.	4401 CS-704	Elective-V*	3	0	0	30	20	50	100	150	3
5.	4405 CS-705	Open Elective	3	0	0	30	20	50	100	150	3
		Practical/Training/Project									
6.	40408 CS-751	*Industrial Training & Seminar	0	0	2	-	20	20	30	50	1
7.	40409 CS-753	Project	0	0	6	-	50	50	150	200	3
8.	40408 GP-701	General Proficiency	-	-	-	-	-	50		50	
		Total	15	0	8					1050	19

Elective III- Choose anyone from the followings

(Cryptography and Network Security, High Speed Network, Android Operating System, Service Oriented Architecture)

Elective IV- Choose anyone from the followings

(Data Mining & Data Warehousing, Distributed Databases, Bio-informatics, Quantum Computing)

Elective-V Choose anyone from the followings

(Mobile Computing, Neural Network, Fuzzy System, Neural Language Processing)

Open Elective-I

(Software Project Management, Software Quality Engineering, Rural Technology & Community Development, Software Testing & Audit)

*Practical training done after 6th Semester would be evaluated in 7th semester through Report and Viva-voce.

**Student has to deliver a seminar on the topic provided by the departmental concern.

***Project has to be initiated in 7th semester beginning and completed by the end of 8th semester with proper report and demonstration.

Year 4th, Semester-VIII

S. No.	Course Code	Course Title/Subjects	Periods			Evaluation Scheme				Subject Total	Credit
						Sessional		Examination			
		Theory	L	T	P	C A	TA	Total			
1.	4407 CS-801	Digital image Processing	3	1	0	30	20	50	100	150	4
2.	4406 CS-802	Elective VI	3	1	0	30	20	50	100	150	4
3.	4408 CS-803	Open Elective-II*	3	1	0	30	20	50	100	150	4
		Practical/Training/Project									
5.	40411 CS-851	Digital image Processing Lab	0	0	2	-	20	20	30	50	1
7.	40412 CS-853	Project	0	0	12	-	100	100	400	500	6
8.	40413 GP-801	General Proficiency	-	-	-	-	-	50		50	
		Total	9	3	14					1050	19

Elective-VI * Chosen anyone from the followings
(Distributed System, Real Time Systems, Pattern Recognition, Cluster Computing, Grid Computing)

Open Elective-II ** Chosen anyone from the followings
(Entrepreneurship Development, Introduction to Bio Technology, Ethical Hacking & Prevention, Non conventional energy resources)

Course Outcomes(CO)

2401 COMPUTER ORGANIZATION

CO1: Define the fundamental components for computer design and organisation.

CO2: Describe the operations and language fetch the register transfer, BUS, Data Representation

CO3: Apply arithmetic microoperations in computer system.

CO4: Analyze cache mapping and Understand the organization of memory and memory management hardware.

CO5: Elaborate advanced concepts of computer architecture, Parallel Processing, interprocessor communication and synchronization.

2402 DATA STRUCTURE USING C

CO1: Understand the concept of Dynamic memory management, data types, algorithms, Big O notation.

CO2: Understand basic data structures such as arrays, linked lists, stacks and queues.

CO3: Describe the hash function and concepts of collision and its resolution methods.

CO4: Solve problem involving graphs, trees and heaps.

CO5: Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data.

2403 CBNST

CO1: Obtain an intuitive and working understanding of numerical methods for the basic problems of numerical analysis.

CO2: Gain experience in the implementation of numerical methods using a computer.

CO3: Trace error in these methods and need to analyze and predict it.

2404 DIGITAL LOGIC DESIGN

CO1: Apply the principles of number system, binary codes and Boolean algebra to minimize logic expressions

CO2: Develop K-maps to minimize and optimize logic functions up to 5 variables

CO3: Acquire knowledge about various logic gates and logic families and analyze basic circuits of these families.

CO4: Design various combinational and sequential circuits such as encoders, decoders and counters using multiplexers, and flip-flops

CO5: Describe and compare various memory systems, shift registers and analog to digital and digital to analog conversion circuits

2405 DISCRETE STRUCTURE

CO1: Define sets, relations, functions; Describe Hasse diagram, Lattice and apply operations on discrete structures in different areas of computing

CO2: Apply mathematical techniques to prove arguments/statement/theorems; define pigeonhole principle and discuss its significance; solve problems using counting techniques and combinatorics

CO3: Describe propositional Logic and test the validity of arguments, analyze finite state machine as model of physical system and language recognizers

CO4: Classify algebraic structures, Groups, Rings and Fields; solve problems using algebraic structures.

CO5: Define graphs; classify graphs by their characteristics, apply graph theory to solve problems in different areas of computer science Mapping of COs and Pos.

2406 Design and Analysis of Algorithm

CO1: Define the basic concepts of algorithms and analyze the performance of algorithms.

CO2: Discuss various algorithm design techniques for developing algorithms.

CO3: Discuss various searching, sorting and graph traversal algorithms.

CO4: Understand NP completeness and identify different NP

2407 Database Management System

CO1:Describe DBMS architecture, physical and logical database designs, database modeling, relational, hierarchical and network models.

CO2: Identify basic database storage structures and access techniques such as file organizations, indexing methods including B tree, and hashing.

CO3: Learn and apply Structured query language (SQL) for database definition and database manipulation.

CO4: Demonstrate an understanding of normalization theory and apply such knowledge to the normalization of a database.

CO5: Understand various transaction processing, concurrency control mechanisms and database protection mechanisms.

2408 Object Oriented System

CO1:Able to develop programs with reusability

CO2: Develop programs for file handling

CO3: Handle exceptions in programming

CO4: Develop applications for a range of problems using object-oriented programming techniques

CO5: Develop java-based projects

2409 Software Engineering

CO1:Plan a software engineering process life cycle , including the specification, design, implementation, and testing of software systems that meet specification, performance, maintenance and quality requirements

CO2: Able to elicit, analyze and specify software requirements through a productive working relationship with various stakeholders of the project

CO3: Analyze and translate a specification into a design, and then realize that design practically, using an appropriate software engineering methodology.

CO4: Know how to develop the code from the design and effectively apply relevant standards and perform testing, and quality management and practice

CO5: Able to use modern engineering tools necessary for software project management, time management and software reuse.

2410 Theory of Automata and Formal Language

CO1: Understand the basic concepts of formal languages, automata and grammar types, as well as the use of formal languages and reduction in normal forms

CO2: Demonstrate the relation between regular expressions, automata, languages and grammar with formal mathematical methods

CO3: Design push down automata, cellular automata and turing machines performing tasks of moderate complexity

CO4: Analyze the syntax and formal properties, parsing of various grammars such as LL(k) and LR(k)

CO5: Describe the rewriting systems and derivation languages

3401 COMPUTER GRAPHICS

CO1: Understand the basics of computer graphics, different graphics systems and applications of computer graphics.

CO2: Discuss various algorithms for scan conversion and filling of line, circle, basic objects and their comparative analysis

CO3: Use of geometric transformations on graphics objects and their application in composite form.

CO4: Extract scene with different clipping methods and its transformation to graphics display device, Understand polygon, segment .

CO5: Explore projections and visible surface detection techniques for display of 3D scene on 2D screen.

3402 COMPILER DESIGN

CO1: Discuss the major phases of compilers and use the knowledge of the Lex tool

CO2: Develop the parsers and experiment with the knowledge of different parsers design without automated tools.

CO3: Describe intermediate code representations using syntax trees and DAG's as well as use this knowledge to generate intermediate code in the form of three address code representations.

CO4: Classify various storage allocation strategies and explain various data structures used in symbol tables

CO5: Summarize various optimization techniques used for dataflow analysis and generate machine code from the source code of a novel language.

3403 Principles of Programming Languages

CO1: Describe distinguishing characteristics of declarative (functional & logical) and imperative (procedural & object-oriented) programming language paradigms and explain how these characteristics manifest in historic and contemporary programming languages.

CO2: Evaluate syntactic, semantic, and pragmatic tradeoffs among the various programming paradigms and programming languages

CO3: Describe the function of Language Processing Components (e.g. Scanner, Parser, etc.)

CO4: Choose an appropriate programming language solution for a given programming task

3404 WEB TECHNOLOGY

CO1: Discuss the insights of internet programming and implement complete application over the web.

CO2: Demonstrate the important HTML tags for designing static pages and separate design from content using Cascading Style sheet.

CO3: Utilize the concepts of JavaScript and Java

CO4: Use web application development software tools i.e. Ajax, PHP and XML etc. and identify the environments currently available on the market to design web sites.

CO5: Develop web based projects using web technologies.

3405 Industrial economics and principles of management

CO1: Able to understand the concept and Importance of Industrial economics.

CO2: Demonstrate the roles, skills and functions of management etc.

CO3: To provides the important tools and techniques to be used in the performance of managerial job.

CO4: To help the students to develop cognizance of the Importance of management principles

3406 Operating System

CO1: Understand the basics of operating systems like kernel, shell, types and views of operating systems

CO2: Describe the various CPU scheduling algorithms and remove deadlocks.

CO3: Explain various memory management techniques and concept of thrashing

CO4: Use disk management and disk scheduling algorithms for better utilization of external memory.

CO5: Recognize file system interface, protection and security mechanisms.

3407 Computer Network

CO1: Understand computer network basics, network architecture, TCP/IP and OSI reference models.

CO2: Identify and understand various techniques and modes of transmission

CO3: Describe data link protocols, multi-channel access protocols and IEEE 802 standards for LAN

CO4: Describe routing and congestion in network layer with routing algorithms and classify IPV4 addressing scheme

CO5: Discuss the elements and protocols of transport layer

3408 Artificial Intelligence

CO1: Differentiate between different knowledge representation techniques and describe methods of knowledge acquisition and extraction.

CO2: Describe various learning and planning techniques for different types of expert systems such as neural, fuzzy and real expert system.

CO3: Develop AI projects using various available tools.

CO4: Analyze the development process of expert system through various case studies.

3409 Operations Research

1. Formulate and solve problems as networks and graphs.
2. Construct linear integer programming models and discuss the solution techniques.
3. Set up decision models and use some solution methods for nonlinear optimization problems.
4. Propose the best strategy using decision making methods under uncertainty and game theory.

3410 Organizational Behavior and principles of management

CO1: To understand the conceptual framework of the discipline of OB and its practical applications in the organizational set up.

CO2: To deeply understand the role of individual, groups and structure in achieving organizational goals effectively and efficiently.

CO3: To critically evaluate and analyze various theories and models that contribute to the overall understanding of the discipline.

CO4: To develop creative and innovative ideas that could positively shape the organizations.

CO5: To accept and embrace working with different people from different cultural and diverse backgrounds in the workplace.

4401 Advance Computer Architecture

CO1: Understand the need for parallel computing, parallel architectural classification schemes, memory hierarchy, quantitative evaluation of performance gain using memory, cache miss/hits.

CO2: Describe Multi-core architectures, program optimizations techniques, building of some of these techniques in compilers, OpenMP and other message passing libraries, threads, mutex etc.

CO3: Develop Multi-threaded architectures and demonstrate the addressing modes used in instructions.

CO4: Synchronization primitives: Atomic primitives; locks: TTS, ticket, array; barriers: central and tree; performance implications in shared memory programs; Chip multiprocessors -cycle, and pipelined implementations of a simple instruction set.

CO5: Apply algorithms for arithmetic operations and Compiler optimization issues ; Data dependence analysis, solving data dependence equations (integer linear programming problem); Loop optimizations; Memory hierarchy issues in code optimization.

4402 Cryptography and network security

CO1:After successful completion of the course, the learners would be able to

CO2: Provide security of the data over the network.

CO3: Do research in the emerging areas of cryptography and network security.

CO4: Implement various networking protocols. CO4: Protect any network from the threats in the world.

4403 Data mining and data warehousing

CO1: Ability to add mining algorithms as a component to the existing tools

CO2: Demonstrate the classification, clustering and etc. in large data sets.

CO3: Ability to apply mining techniques for realistic data.

CO4: Interpret different algorithms standards used in research.

4404 Neural Networks

CO1:Analyze the concept of neural networks for learning linear and non-linear activation functions

CO2: Identify and describe Fuzzy Logic and Artificial Neural Network techniques in building intelligent machines.

CO3: Apply Artificial Neural Network & Fuzzy Logic models to handle uncertainty and solve engineering problems.

CO4: Recognize the feasibility of applying a Neuro-Fuzzy model for a particular problem.

CO5: Integrate neural network and fuzzy logic to extend the capabilities for efficient and effective problem solving methodologies.

4405 Software project management

CO1: Able to prepare SRS document, design document, test cases and software configuration management and risk management related document.

CO2: Develop function oriented and object oriented software design using tools like rational rose.

CO3: Able to perform unit testing and integration testing.

CO4: Apply various white box and black box testing techniques

4406 Digital Image Processing

CO1: Review the fundamental concepts of a digital image processing system.

CO2: Analyze images in the frequency domain using various transforms.

CO3: Evaluate the techniques for image enhancement and image restoration.

CO4: Categorize various compression techniques.

CO5: Interpret Image compression standards.

4407 Elective VI : Distributed System

CO1: Define the characterization of Distributed Systems, Theoretical Foundation for Distributed System and Concepts in Message Passing Systems.

CO2: Explain the Distributed Mutual Exclusion and Distributed Deadlock Detection.

CO3: Apply the Agreement Protocols and Distributed Resource Management.

CO4: Analyze the Failure Recovery in Distributed Systems and Fault Tolerance.

CO5: Evaluate the Transactions and Concurrency Control, Distributed Transactions and Replication.

4408 OPEN Elective -II : Entrepreneurship Development

CO1: Review the fundamental concepts of EDP.

CO2: Recognize the importance of environment and Entrepreneurship Development

CO3: Analyze interaction between social and Entrepreneurship Development processes.

CO4: Conceptualize the processes and various factors involved in the formation of EDP.

COMPUTER ORGANISATION

Unit I

Register Transfer Language, Bus and Memory Transfers ,Bus Architecture, Bus Arbitration ,Arithmetic Logic, Shift Microoperation, Arithmetic Logic Shift Unit Design of Fast address, Arithmetic Algorithms (addition, subtraction, Booth Multiplication) IEEE standard for Floating point numbers.

Unit II

Hardwired & Micro Programmed (Control Unit): Fundamental Concepts (Register Transfers microinstruction with Next-address field, Prefetching Microinstruction). Storing a word in memory), Execution of a complete instruction,

Multiple-Bus organization, Hardwired Control, Micro programmed control(Microinstruction) Microprogram sequencing, Wide-Branch addressing Performing of arithmetic or logical operations, Fetching a word from memory,

Unit III

Processor Organization: General register organization ,Stack organization Addressing mode,Instruction format, Data transfer & manipulations Program Control, Reduced Instruction Set Computer

Unit IV

I/O Interface Modes of transfer, Interrupts & Interrupt handling Direct Memory access Input-Output processor, Serial Communication.

Unit V

Memory Hierarchy Main Memory (RAM and ROM Chips) Organization of 2D and 2^{1/2}D, Auxiliary memory, Cache memory Virtual Memory Memory management hardware Basic of Pipelining , throughput and speedup, pipeline hazard Basics of parallel processor, concurrent access to memory and cache memory

Text Books:

1. Computer System Architecture, M. Mano(PHI)

Reference Books:

1. Computer Organization, Vravice, Zaky & Hamacher (TMH Publication)
2. Structured Computer Organization, Tannenbaum(PHI)
3. Computer Organization, Stallings(PHI)
4. Computer Organization, John P.Hayes (McGraw Hill)

DATA STRUCTURES USING - C

Unit - I

Introduction: Basic Terminology, Elementary Data Organization, Structure operations, Algorithm Complexity and Time-Space trade-off

Arrays: Array Definition, Representation and Analysis, Single and Multidimensional Arrays, address calculation, application of arrays, Array as Parameters, Ordered List, Sparse Matrices and Vectors.

Stacks: Array Representation and Implementation of stack, Operations on Stacks: Push & Pop, Array Representation of Stack, Linked Representation of Stack, Operations Associated with Stacks, Application of stack: Conversion of Infix to Prefix and Postfix Expressions, Evaluation of postfix expression using stack.

Recursion: Recursive definition and processes, recursion in C, example of recursion, Tower of Hanoi Problem, simulating recursion, Backtracking, recursive algorithms, principles of recursion, tail recursion, removal of recursion.

UNIT - II

Queues: Array and linked representation and implementation of queues, Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, D-queues and Priority Queues.

Linked list: Representation and Implementation of Singly Linked Lists, Two-way Header List, Traversing and Searching of Linked List, Overflow and Underflow, Insertion and deletion to/from Linked Lists, Insertion and deletion Algorithms, Doubly linked list, Linked List in Array, Polynomial representation and addition, Generalized linked list, Garbage Collection and Compaction.

UNIT – III

Trees: Basic terminology, Binary Trees, Binary tree representation, algebraic Expressions, Complete Binary Tree, Extended Binary Trees, Array and Linked Representation of Binary trees, Traversing Binary trees, Threaded Binary trees, Traversing Threaded Binary trees, Huffman algorithm.

Searching and Hashing: Sequential search, binary search, comparison and analysis, Hash Table, Hash Functions, Collision Resolution Strategies, Hash Table Implementation.

UNIT – IV

Sorting: Insertion Sort, Bubble Sorting, Quick Sort, Two Way Merge Sort, Heap Sort, Sorting on Different Keys, Practical consideration for Internal Sorting.

Binary Search Trees: Binary Search Tree (BST), Insertion and Deletion in BST, Complexity of Search Algorithm, Path Length, AVL Trees, B-trees.

UNIT - V

Graphs: Terminology & Representations, Graphs & Multi-graphs, Directed Graphs, Sequential Representations of Graphs, Adjacency Matrices, Traversal, Connected Component and Spanning Trees, Minimum Cost Spanning Trees.

File Structures: Physical Storage Media File Organization, Organization of records into Blocks, Sequential Files, Indexing and Hashing, Primary indices, Secondary indices, B+ Tree index Files, B Tree index Files, Indexing and Hashing Comparisons.

Reference text books:

1. Horowitz and Sahani, “Fundamentals of data Structures”, Galgotia Publication Pvt. Ltd., New Delhi.
2. R. Kruse et al, “Data Structures and Program Design in C”, Pearson Education Asia, Delhi-2002
3. A. M. Tenenbaum, “Data Structures using C & C++”, Prentice-Hall of India Pvt. Ltd., New Delhi.

Supplementary reference books:

1. K Loudon, “Mastering Algorithms With C”, Shroff Publisher & Distributors Pvt. Ltd.
2. Bruno R Preiss, “Data Structures and Algorithms with Object Oriented Design Pattern in C++”, Jhon Wiley & Sons, Inc.
3. Adam Drozdek, “Data Structures and Algorithms in C++”, Thomson Asia Pvt. Ltd.(Singapore)

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Unit-I

Introduction: Numbers and their accuracy, Computer Arithmetic, Mathematical preliminaries, Errors and their Computation, General error formula, Error in a series approximation

Solution of Algebraic and Transcendental Equation:

Bisection Method, Iteration method, Method of false position, Newton-Raphson method, Methods of finding complex roots, Muller's method, Rate of convergence of Iterative methods, Polynomial Equations.

Unit-II

Interpolation: Finite Differences, Difference tables. Polynomial Interpolation: Newton's forward and backward formula.

Central Difference Formulae: Gauss forward and backward formula, Sterling's, Bessel's, Everett's formula. Interpolation with unequal intervals: Lagrange's Interpolation, Newton Divided difference formula, Hermit's Interpolation,

Unit-III

Numerical Integration and Differentiation: Introduction, Numerical differentiation Numerical Integration:

Trapezoidal rule, Simpson's $1/3$ and $3/8$ rule, Boole's rule, Waddle's rule.

Unit-IV

Solution of differential Equations: Picard's Method, Euler's Method, Taylor's Method, Runge-Kutta Methods, Predictor Corrector Methods, Automatic Error Monitoring and Stability of solution

Unit-V

Statistical Computation: Frequency chart, Curve fitting by method of least squares, fitting of straight lines, polynomials, exponential curves etc, Data fitting with Cubic splines, Regression Analysis, Linear and Non linear Regression, Multiple regression, Statistical Quality Control methods.

References:

1. Rajaraman V, "Computer Oriented Numerical Methods", Pearson Education
2. Gerald & Whealey, "Applied Numerical Analyses", AW
3. Jain, Iyengar and Jain, "Numerical Methods for Scientific and Engineering Computations", New Age Int.
4. Grewal B S, "Numerical methods in Engineering and Science", Khanna Publishers, Delhi
5. T Veerarajan, T Ramachandran, "Theory and Problems in Numerical Methods, TMH
6. Pradip Niyogi, "Numerical Analysis and Algorithms", TMH
7. Francis Scheld, "Numerical Analysis", TMH
8. Sastry S. S, "Introductory Methods of Numerical Analysis", Pearson Education.
9. Gupta C.B., Vijay Gupta, "Introduction to Statistical Methods", Vikas Publishing.
10. Goyal, M, "Computer Based Numerical and Statistical Techniques", Firewall Media, New Delhi.

DIGITAL LOGIC DESIGN

Unit-I : Introduction

Characteristics of digital system, Types of Digital circuits, Number system: Direct conversion between bases Negative numbers & BCD and their arithmetic's, Boolean Algebra, Minimization of Boolean Functions :Map & Tabular method upto 6 variable and multiple output circuits Error detecting

& correcting codes, Hamming & cyclic codes.

6

Unit-II : Combinational Logic Circuits

Design Procedure, Adders, subtractors & code conversion, Multiplexers/ Demultiplexers, encoder decoders, decimal adders & amplitude comparators, ROM as decoder, PLA & PAL.

6

Unit-III : Sequential Logic Circuits

Flip –Flops and their conversions, Analysis and synthesis of synchronous sequential circuit, excitation table, state table & state diagram. Design of synchronous counters, shift registers and their applications. **Algorithm State Machine:** ASM chart, Timing considerations, Control Implementation Design with Multiplexers, PLA control

Asynchronous Sequential Circuits: Analysis Procedure Reduction of state & flow table, Race free

state assignment.

10

Unit-IV : Logic Families

Diode, BJT & MOS as a switching element concept of transfer characteristics, Input characteristics and output characteristics of logic gates, Fan-in, Fan-out, Noise margin, circuit concept and comparison of various logic families: TTL, IIL, ECL, NMOS, CMOS Tri-state logic, open collector output, Interfacing between logic families, packing density, power consumption & gate delay.

8

Unit-V : Hazard and Fault Detection

Static and dynamic Hazard : Gate delay, Generation of spikes, Determination of hazard in

combinational circuits, Fault detection methods: Fault Table & Path sensitizing methods.

5

Unit-VI : Memories

Sequential, Random Access, NMOS & CMOS Static and Dynamic Memory elements, one and multi-dimensional selection arrangement, Read-only memories, Formation of memory banks.

5

Text Books :

1. Digital Design by M Moris Mano, 2nd Edn.PHI
2. Introduction to Digital Microelectronic Circuits, by Gopalan, TMH

Reference Books :

1. Switching Circuit & Logic Design by Hill & Peterson, Wiley
2. Digital Circuit & Logic Design, by Holsworth.

DISCRETE STRUCTURES

Unit-I:

Set Theory: Definition of sets, countable and uncountable sets, Venn Diagrams, proofs of some general identities on sets

Relation: Definition, types of relation, composition of relations, Pictorial representation of relation, equivalence relation, partial ordering relation.

Function: Definition, type of functions, one to one, into and onto function, inverse function, composition of functions, recursively defined functions.

Theorem proving Techniques: mathematical induction (simple and strong), pigeonhole principle, prove by contradiction.

Unit-II:

Algebraic Structures: Definition, Properties, types: Semi Groups, Monoid, Groups, Abelian group, properties of groups, Subgroup, cyclic groups, Cosets, factor group, Permutation groups, Normal subgroup, Homomorphism and isomorphism of Groups, example and standard results, Rings and Fields: definition and standard results.

Unit-III:

Posets, Hasse Diagram and Lattices: Introduction, ordered set, Hasse diagram of partially, ordered set, isomorphic ordered set, well ordered set, properties of Lattices, bounded I and complemented lattices.

Boolean Algebra: Basic definitions, sum of products and product of sums, form in Boolean Algebra, Logic gates and Karnaugh maps.

Tree: Definition, Rooted tree, properties of trees, binary search tree, tree traversal.

Unit-IV:

Propositional Logic: Proposition, First order logic, Basic logical operation, truth tables, tautologies, Contradictions, Algebra of Proposition, logical implications, logical equivalence, predicates, Universal and existential quantifiers.

Unit-V:

Combinatorics & Graphs: Recurrence Relation, Generating function., Simple graph, multi graph, graph terminology, representation of graphs, Bipartite, Regular, Planar and connected graphs, connected components in a graph, Euler graphs, Hamiltonian path and circuits, Graph coloring, chromatic number, isomorphism and Homomorphism of graphs.

Text books and Supplementary reading:

1. Liptschutz, Seymour, “ Discrete Mathematics”, McGraw Hill.
2. Trembley, J.P & R. Manohar, “Discrete Mathematical Structure with Application to Computer Science”, McGraw Hill.
3. Kenneth H. Rosen, “Discrete Mathematics and its applications”, McGraw Hill.
4. Deo, Narsingh, “Graph Theory With application to Engineering and Computer.Science.”, PHI.
5. Krishnamurthy, V., “Combinatorics Theory & Application”, East-West Press Pvt. Ltd., New Delhi.

COMPUTER ORGANIZATION LAB

1. Bread Board Implementation of Flip-Flops.
2. Experiments with clocked Flip-Flop.
3. Design of Counters.
4. Bread Board implementation of counters & shift registers.
5. Implementation of Arithmetic algorithms.
6. Bread Board implementation of Adder/Subtractor (Half, Full)
7. Bread Board implementation of Binary Adder.
8. Bread Board implementation of Seven Segment Display.

***Institute may also develop the experiment based on the infrastructure available with them.**

DATA STRUCTURES LAB

Write Program in C or C++ for following.

- 1 Array implementation of Stack, Queue, Circular Queue, List.
- 2 Implementation of Stack, Queue, Circular Queue, List using Dynamic memory Allocation.
- 3 Implementation of Tree Structures, Binary Tree, Tree Traversal, Binary Search Tree, Insertion and
Deletion in BST.
- 4 Implementation of Searching and Sorting Algorithms.
- 5 Graph Implementation, BFS, DFS, Min. cost spanning tree, shortest path algorithm.

NUMERICAL TECHNIQUES LAB

1. To deduce error involved in polynomial equation.
2. To Find out the root of the Algebraic and Transcendental equations using Bisection, Regula-falsi, Newton Raphson and Iterative Methods. Also give the rate of convergence of roots in tabular form for each of these methods.
3. To implement Newton's Forward and Backward Interpolation formula.
4. To implement Gauss Forward and Backward, Bessel's, Sterling's and Evertt's Interpolation formula
5. To implement Newton's Divided Difference and Langranges Interpolation formula.
6. To implement Numerical Differentiations.
7. To implement Numerical Integration using Trapezoidal, Simpson 1/3 and Simpson 3/8 rule.

8. To implement Least Square Method for curve fitting.
9. To draw frequency chart like histogram, frequency curve and pie-chart etc.
10. To estimate regression equation from sampled data and evaluate values of standard deviation, t-statistics, regression coefficient, value of R^2 for atleast two independent variables.

DIGITAL ELECTRONICS LAB

1. Bread-board implementation of various flip-flops.
2. Bread-board implementation of counters & shift registers.
3. Determination of Delay time and NAND, NOR, Ex-OR, AND & OR Gates.
4. Transfer characteristics of TTL inverters & TTL Schmitt Trigger inverter.
5. Transfer characteristics of CMOS inverters series and CD40 series and estimation of Gate delay of CD40 series CMOS inverter.
6. Monoshot multivibrators using 74121 and 74123.
7. Clock circuit realization using 555 and CMOS inverter and quartz crystal.
8. Adder/ subtractor operation using IC7483 4 bit/ 8 bit.
9. Demultiplexer / Decoder operation using IC-74138.
10. Modulo N counter using programmable counter 74190.

DESIGN & ANALYSIS OF ALGORITHMS

Unit 1:

Introduction: Characteristics of algorithm. Analysis of algorithm: Asymptotic analysis of complexity bounds-best, average and worst-case behavior, Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

Unit 2:

Fundamental Algorithmic Strategies: Brute-Force, Greedy, Dynamic Programming. Branch-and-Bound and Backtracking methodologies for the design of algorithms, Illustrations of these techniques for Problem-Solving, Bin Packing, Knap Sack TSP. Heuristics - characteristics and their application domains.

Unit 3:

Graph and Tree Algorithms: Traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); Shortest path algorithms, Transitive closure, Minimum Spanning Tree, Topological sorting. Network Flow Algorithm.

Unit 4:

Tractable and Intractable Problems: Computability of Algorithms, Computability classes-P, NP, NP-complete and NP-hard. Cook's theorem, Standard NP-complete problems and Reduction techniques.

Unit 5:

Advance Topics: Approximation of Algorithms, Randomized Algorithms, Class of Problems Beyond NP-P Space.

References:

1. Cormen, Rivest, Lisserson,: "Algorithm", PHI.
2. Basse, "Computer Algorithms: Introduction to Design & Analysis", Addison Wesley.
3. Horowitz & Sahani, "Fundamental of Computer Algorithm", Galgotia.

DATABASE MANAGEMENT SYSTEM

Unit- I

Introduction: An overview of database management system, database system Vs file system, Database system concepts and architecture, data models schema and instances, data independence and data base language and interfaces, Data definitions language, DML, Overall Database Structure.

Data Modeling using the Entity Relationship Model:

ER model concepts, notation for ER diagram, mapping constraints, keys, Concepts of Super Key, candidate key, primary key, Generalization, aggregation, reduction of an ER diagrams to tables, extended ER model, relationships of higher degree.

Unit- II

Relational data Model and Language: Relational data model concepts, integrity constraints: entity integrity, referential integrity, Keys constraints, Domain constraints, relational algebra, relational calculus, tuple and domain calculus,

Introduction to SQL: Characteristics of SQL. Advantage of SQL. SQL data types and literals. Types of SQL commands. SQL operators and their procedure. Tables, views and indexes. Queries and sub queries. Aggregate functions. Insert, update and delete operations. Joins, Unions, Intersection, Minus, Cursors in SQL.

Unit- III

Data Base Design & Normalization:

Functional dependencies, normal forms, first, second, third normal forms, BCNF, inclusion dependences, loss less join decompositions, normalization using FD, MVD, and JDs, alternative approaches to database design.

Unit- IV

Transaction Processing Concepts: Transaction system, Testing of serializability, Serializability of schedules, conflict & view serializable schedule, recoverability, Recovery from transaction failures, log based recovery, checkpoints, deadlock handling.

Unit- V

Concurrency Control Techniques: Concurrency control, locking Techniques for concurrency control, Time stamping protocols for concurrency control, validation based protocol, multiple granularity, Multi version schemes, Recovery with concurrent transaction.

Text Books

- 1 Date C J, "An Introduction To Database System", Addison Wesley
- 2 Korth, Silbertz, Sudarshan, "Database Concepts", McGraw Hill
- 3 Elmasri, Navathe, "Fundamentals Of Database Systems", Addison Wesley
- 4 Leon & Leon, "Database Management System", Vikas Publishing House.

References

- 1 Bipin C. Desai, "An introduction to Database Systems", Galgotia Publication
- 2 Majumdar & Bhattacharya, "Database Management System", TMH
- 3 Ramakrishnan, Gehrke, "Database Management System", McGraw Hill
4. Kroenke, "Database Processing: Fundamentals, Design and Implementation", Pearson Education.
5. Maheshwari Jain, "DBMS: Complete Practical Approach", Firewall Media, New Delhi.

OBJECT ORIENTED SYSTEMS

Unit – I

Object Modeling: Objects and classes, links and association, generalization and inheritance, aggregation, abstract class, multiple inheritance, meta data, candidate keys, constraints.

Unit – II

Dynamic Modeling: Events and states, operations, nested state diagrams and concurrency, advanced dynamic modeling concepts, a sample dynamic model.

Unit – III

Functional Modeling: Data flow diagram, specifying operations, constraints, a sample functional model. OMT (object modeling techniques) methodologies, examples and case studies to demonstrate methodologies, comparisons of methodologies, SA/SD, JSD.

Unit – IV

Java Programming: Introduction, Operator, Data types, Variables, Methods & Classes, Multithread Programming, I/O, Java Applet.

Java Library: String Handling, Input/Output exploring Java.io, Networking, Applets classes, Event Handling, Introduction to AWT, Working with window, Graphics, AWT Controls, Layout Manager and Menus, Images, Additional packages.

Unit – V

Software Development using Java:

Java Beans, Java Swing, Java Servlets, Migrating from C++ to java, Application of java, Dynamic Billboard Applet, Image Menu: An image based menu, Lavatron Applets, Scrabblets, JDBC, Brief functioning of upper layer E-mail and their applications.

Text Books:

1. James Rumbaugh et al, “Object Oriented Modeling and Design”, PHI

2. Herbert Schildt, “The Complete Reference: Java”, TMH.
3. E. Balagurusamy, “Programming in JAVA”, TMH.

References:

1. Booch Grady, “Object Oriented Analysis & Design with application 3/e”, Pearson Education, New Delhi.
2. Bjarne Stroustrup, “C++ Programming Language”, Addison Wesley
3. E. Balagurusamy, “Object Oriented Programming with C++”, TMH.

SOFTWARE ENGINEERING

Unit-I: Introduction

Introduction to Software Engineering, Software Components, Software Characteristics, Software Crisis, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes, Software Quality Attributes.

Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Models.

Unit-II: Software Requirement Specifications (SRS)

Requirement Engineering Process: Elicitation, Analysis, Documentation, Review and Management of User Needs, Feasibility Study, Information Modeling, Data Flow Diagrams, Entity Relationship Diagrams, Decision Tables, SRS Document, IEEE Standards for SRS.

Software Quality Assurance (SQA): Verification and Validation, SQA Plans, Software Quality Frameworks, ISO 9000 Models, SEI-CMM Model.

Unit-III: Software Design

Basic Concept of Software Design, Architectural Design, Low Level Design: Modularization, Design Structure Charts, Pseudo Codes, Flow Charts, Coupling and Cohesion Measures, Design Strategies: Function Oriented Design, Object Oriented Design, Top-Down and Bottom-Up Design.

Software Measurement and Metrics: Various Size Oriented Measures: Halstead's Software Science,

Function Point (FP) Based Measures, Cyclomatic Complexity Measures: Control Flow Graphs.

Unit-IV: Software Testing

Testing Objectives, Unit Testing, Integration Testing, Acceptance Testing, Regression Testing, Testing for Functionality and Testing for Performance, Top-Down and Bottom-Up Testing Strategies: Test Drivers and Test Stubs, Structural Testing (White Box Testing), Functional Testing (Black Box Testing), Test Data Suit Preparation, Alpha and Beta Testing of Products.

Static Testing Strategies: Formal Technical Reviews (Peer Reviews), Walk Through, Code Inspection, Compliance with Design and Coding Standards.

Unit-V: Software Maintenance and Software Project Management

Software as an Evolutionary Entity, Need for Maintenance, Categories of Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Software Re-Engineering, Reverse Engineering. Software Configuration Management Activities, Change Control Process, Software Version Control, An Overview of CASE Tools.

Estimation of Various Parameters such as Cost, Efforts, Schedule/Duration, Constructive Cost Models (COCOMO), Resource Allocation Models, Software Risk Analysis and Management.

Reference Books:

1. R. S. Pressman, Software Engineering: A Practitioners Approach, McGraw Hill.
2. Rajib Mall, Fundamentals of Software Engineering, PHI Publication.
3. K. K. Aggarwal and Yogesh Singh, Software Engineering, New Age International Publishers.
4. Carlo Ghezzi, M. Jarayeri, D. Manodrioli, Fundamentals of Software Engineering, PHI Publication.
5. Ian Sommerville, Software Engineering, Addison Wesley.
6. Pankaj Jalote, Software Engineering, Narosa Publication
7. Pfleeger, Software Engineering, Macmillan Publication.
8. A. Leon and M. Leon, Fundamentals of Software Engineering, Vikas Publication.

THEORY OF AUTOMATA & FORMAL LANGUAGES

Unit – I

Introduction to defining language, Kleene closures, Arithmetic expressions, defining grammar, Chomsky hierarchy, Finite Automata (FA), Transition graph, generalized transition graph.

Unit – II

Nondeterministic finite Automata (NFA), Deterministic finite Automata (DFA), Construction of DFA from NFA and optimization, FA with output: Moore machine, Mealy machine and Equivalence, Applications and Limitation of FA

Unit – III

Arden Theorem, Pumping Lemma for regular expressions, Myhill-Nerode theorem, Context free grammar: Ambiguity, Simplification of CFGs, Normal forms for CFGs, Pumping lemma for CFLs, Decidability of CFGs, Ambiguous to Unambiguous CFG.

Unit – IV

Push Down Automata (PDA): Description and definition, Working of PDA, Acceptance of a string by PDA, PDA and CFG, Introduction to auxiliary PDA and Two stack PDA.

Unit – V

Turing machines (TM): Basic model, definition and representation, Language acceptance by TM, TM and Type – 0 grammar, Halting problem of TM, Modifications in TM, Universal TM, Properties of recursive and recursively enumerable languages, unsolvable decision problem, undecidability of Post correspondence problem, Church's Thesis, Recursive function theory, Godel Numbering.

Text Books and References:

1. Hopcroft, Ullman, "Introduction to Automata Theory, Language and Computation", Nerosa

Publishing House

2. K.L.P. Mishra and N.Chandrasekaran, "Theory of Computer Science(Automata, Languages and Computation)", PHI
3. Martin J. C., "Introduction to Languages and Theory of Computations", TMH
4. Papadimitrou, C. and Lewis, C.L., "Elements of theory of Computations", PHI
5. Cohen D. I. A., "Introduction to Computer theory", John Wiley & Sons
6. Kumar Rajendra, "Theory of Automata (Languages and Computation)", PPM

DESIGN AND ANALYSIS OF ALGORITHMS LAB

Programming assign to each algorithmic strategy:

1. Divide and conquer method (quick sort, merge sort, Strassen's matrix multiplication).
2. Greedy method (knapsack problem, job sequencing, optimal merge patterns, minimal spanning trees).
3. Dynamic programming (multistage graphs, OBST, 0/1 knapsack, traveling salesperson problem).
4. Back tracking (n-queens problem, graph coloring problem, Hamiltonian cycles).
5. Sorting: Insertion sort, Bubble sort and Heap sort.
6. Searching: Sequential and binary search.
7. Selection: Minimum / maximum, kth smallest element.

Java Programming LAB

1. Write a java program to print "Hello"
2. Write a java program that reads your name from command line argument & print Hello *name*.
3. Input any integer n through keyboard and display all prime numbers between 1 and n.
4. Write a java program to generate fibonacci series.
5. Write a program to test command line arguments. Display total number of arguments and also display one argument in one line.
6. Write a Java program to input any integer n through command line argument and display ODD numbers between 1 and n as well as display EVEN numbers between 1 and n
7. Write a java program to simulate a simple calculator. Provide 2 Nos. and an operator as a command line argument to perform the operation of addition, subtraction, multiplication, division, modulo, power. If the operator is not a valid, give appropriate message.
8. Write a java program to print the table of 11 to 20.

9. Create a class Matrix to represent $n \times m$ matrix. Add methods for matrix addition, subtraction, multiplication.
10. Write a java class to store countries and their capitals. Write functions to add new country detail, to change the capital of a country, given a country name find capital and vice versa.
11. Create an applet to draw our national flag.
12. Write a java application that has three Buttons labeled Red, Green, Blue & on clicking the background color should change accordingly & have a label displaying the color selected.
13. Write a program to count the total no. of chars, words, lines, alphabets, digits, white spaces of a given file. Before counting, check whether the file really exists or not. if exists, is it readable? Also give file size.
14. Write an application that creates and starts three threads. Each thread is instantiated from the same class. It executes a loop with 10 iterations. Each iteration displays string "HELLO", sleeps for 300 milliseconds. The application waits for all the threads to complete & displays the message "Good Bye...".
15. Write an application that executes two threads. One thread displays "Good Morning" every 1000 milliseconds & another thread displays "Good Afternoon" every 3000 milliseconds. Create the threads by implementing the Runnable interface.

Software Engineering LAB

1. To study about Software Characteristics.
2. To study about various kinds of software applications.
3. Explain various Software myths.
4. To study about various kinds of Risks.
5. To study about Risk assessment.
6. To study about Risk mitigation(RMMM)
7. To study about requirement specification.
8. To study about Requirement Engineering.
9. To study about System Modeling.
10. To study about architecture of analysis
11. To study about Software Design
12. To study about various kinds of testing techniques.
13. To study about object oriented analysis and design
14. Implementation of one project using software engineering.

DBMS LAB

The Queries to be implemented on DBMS by using SQL.

1. Write the queries for Data Definition and Data Manipulation language.

2. Write SQL queries using Logical operators (=,<,>,etc.).
3. Write SQL queries using SQL operators (Between.... AND, IN(List), Like, ISNULL and also with negating expressions).
4. Write SQL query using character, number, date and group functions.
5. Write SQL queries for Relational Algebra (UNION, INTERSECT, and MINUS, etc.).
6. Write SQL queries for extracting data from more than one table (Equi-Join, Non-Equi-Join , Outer Join)
7. Write SQL queries for sub queries , nested queries.
8. Write programs by the use of PL/SQL.
9. Concepts for ROLL BACK, COMMIT & CHECK POINTS.
10. Create VIEWS, CURSORS, and TRIGGRS & write ASSERTIONS.
11. Create FORMS and REPORTS .

COMPUTER GRAPHICS

Unit-I

Line generation: Points lines, Planes, Pixels and Frame buffers, vector and character generation.

Graphics Primitives: Display devices, Primitive devices, Display File Structure, Display control text.

Unit-II

Polygon: Polygon Representation, Entering polygons, Filling polygons. Segments: Segments table, creating deleting and renaming segments, visibility, image transformations.

Unit-III

Transformations: Matrices transformation, transformation routines, displays procedure. Windowing and

Clipping: Viewing transformation and clipping, generalize clipping, multiple windowing.

Unit-IV

Three Dimension: 3-D geometry primitives, transformations, projection clipping. Interaction: Hardware input devices handling algorithms, Event handling echoing, Interactive techniques.

Unit-V

Hidden Line and Surface: Back face removal algorithms, hidden line methods. Rendering and Illumination: Introduction to curve generation, Bezier, Hermite and B-spline algorithms and their comparisons.

References :

1. Rogers, "Procedural Elements of Computer Graphics", McGraw Hill
2. Asthana, Sinha, "Computer Graphics", Addison Wesley Newman and Sproul, "Principle of Interactive Computer Graphics", McGraw Hill
3. Steven Harrington, "Computer Graphics", A Programming Approach, 2nd Edition

4. Rogar and Adams, "Mathematical Elements of Computer Graphics", McGraw Hill.

COMPILER DESIGN

Unit-I

Introduction to Compiler, Phases and passes, Bootstrapping, Finite state machines and regular expressions and their applications to lexical analysis, Implementation of lexical analyzers, lexical-analyzer generator, LEX-compiler, Formal grammars and their application to syntax analysis, BNF notation, ambiguity, YACC. The syntactic specification of programming languages: Context free grammars, derivation and parse trees, capabilities of CFG.

Unit-II

Basic Parsing Techniques: Parsers, Shift reduce parsing, operator precedence parsing, top down parsing, predictive parsers Automatic Construction of efficient Parsers: LR parsers, the canonical Collection of LR(0) items, constructing SLR parsing tables, constructing Canonical LR parsing tables, Constructing LALR parsing tables, using ambiguous grammars, an automatic parser generator, implementation of LR parsing tables, constructing LALR sets of items.

Unit-III

Syntax-directed Translation: Syntax-directed Translation schemes, Implementation of Syntax-directed Translators, Intermediate code, postfix notation, Parse trees & syntax trees, three address code, quadruple & triples, translation of assignment statements, Boolean expressions, statements that alter the flow of control, postfix translation, translation with a top down parser. More about translation: Array references in arithmetic expressions, procedures call, declarations, case statements.

Unit-IV

Symbol Tables: Data structure for symbols tables, representing scope information. Run-Time

Administration: Implementation of simple stack allocation scheme, storage allocation in block structured

language. Error Detection & Recovery: Lexical Phase errors, syntactic phase errors semantic errors.

Unit-V

Introduction to code optimization: Loop optimization, the DAG representation of basic blocks, value numbers and algebraic laws, Global Data-Flow analysis.

Textbook:

Aho, Sethi & Ullman, "Compiler Design", Addison Wesley.

PRINCIPLES OF PROGRAMMING LANGUAGES

Unit -I

Introduction: Characteristics of programming Languages, Factors influencing the evolution of programming language, developments in programming methodologies, desirable features and design issues. Programming language processors: Structure and operations of translators, software simulated computer, syntax, semantics, structure, virtual computers, binding and binding time.

Unit -II

Elementary and Structured Data Types: Data object variables, constants, data types, elementary data types, declaration, assignment and initialization, enumeration, characters, strings. Structured data type and objects: Specification of data structured types, vectors and arrays, records, variable size data structure, pointers and programmer constructed data structure, Sets files. Sub Program and programmer defined data types: Evolution of data types, abstractions, encapsulations, information hiding, sub programmes, abstract data types.

Unit -III

Sequence Control; Implicit and Explicit sequence control, sequence control with within expression and statements, recursive sub programmes, exception handling, co routines, Scheduled sub programmes, concurrent execution. Data control referencing environments, static and dynamic scope, local data local data referencing environment, shared data: Explicit common environment dynamic scope parameter passing mechanism.

Unit -IV

Storage Management: Major run time requirements, storage management phases, static storage management, stack based, heap based storage management. Syntax and translation: General syntactic criteria, syntactic element of a language, stages in translation, formal syntax and semantics.

Unit -V

Operating and Programming Environment: Batch Processing Environments, Embedded system requirements, Theoretical models, Introduction to Functional Programming, Lambda calculus, Data flow language and Object Oriented language, Comparison in various general and special purpose programming languages e.g. Fortran, C, Pascal, Lisp, etc.

References:

1. Terrance W Pratt, "Programming Languages: Design and Implementation" PHI
2. Sebesta, "Concept of Programming Language", Addison Wesley
3. E Horowitz, "Programming Languages", 2nd Edition, Addison Wesley
4. "Fundamentals of Programming Languages", Galgotia.

WEB TECHNOLOGY

UNIT I: Introduction and Web Development Strategies

History of Web, Protocols governing Web, Creating Websites for individual and Corporate World, Cyber Laws, Web Applications, Writing Web Projects, Identification of Objects, Target Users, Web Team, Planning and Process Development.

UNIT II: HTML, XML and Scripting

List, Tables, Images, Forms, Frames, CSS Document type definition, XML schemes, Object Models, Presenting XML, Using XML Processors: DOM and SAX, Introduction to Java Script, Object in Java Script, Dynamic HTML with Java Script.

UNIT III: Java Beans and Web Servers

Introduction to Java Beans, Advantage, Properties, BDK, Introduction to EJB, Java Beans API.

Introduction to Servlets, Lifecycle, JSDK, Servlet API, Servlet Packages: HTTP package, Working with Http request and response, Security Issues.

UNIT IV: JSP

Introduction to JSP, JSP processing, JSP Application Design, Tomcat Server, Implicit JSP objects, Conditional Processing, Declaring variables and methods, Error Handling and Debugging, Sharing data between JSP pages- Sharing Session and Application Data.

UNIT V: Database Connectivity

Database Programming using JDBC, Studying Javax.sql.*package, accessing a database from a JSP page, Application-specific Database Action, Developing Java Beans in a JSP page, Introduction to Struts framework.

Textbooks:

1. Burdman, "Collaborative Web Development" Addison Wesley.

2. Chris Bates, "Web Programing Building Internet Applications", 2nd Edition, WILEY, Dreamtech
3. Joel Sklar , "Principal of web Design" Vikash and Thomas Learning
4. Horstmann, "CoreJava", Addison Wesley.
5. Herbert Schieldt, "The Complete Reference:Java", TMH.
6. Hans Bergsten, "Java Server Pages", SPD O'Reilly

DATA COMPRESSION

Unit - I:

Compression Techniques: Loss less compression, Lossy Compression, Measures of performance, Modeling and coding, Mathematical Preliminaries for Lossless compression: A brief introduction to information theory, Models: Physical models, Probability models, Markov models, composite source model, Coding: uniquely decodable codes, Prefix codes.

Unit – II:

The Huffman coding algorithm: Minimum variance Huffman codes, Adaptive Huffman coding: Update procedure, Encoding procedure, Decoding procedure. Golomb codes, Rice codes, Tunstall codes, Applications of Hoffman coding: Loss less image compression, Text compression, Audio Compression.

Unit-III:

Coding a sequence, Generating a binary code, Comparison of Binary and Huffman coding, Applications: Bi-level image compression-The JBIG standard, JBIG2, Image compression. Dictionary Techniques: Introduction, Static Dictionary: Diagram Coding, Adaptive Dictionary. The LZ77 Approach, The LZ78 Approach, Applications: File Compression-UNIX compress, Image Compression: The Graphics Interchange Format (GIF), Compression over Modems: V.42 bits, Predictive Coding: Prediction with Partial match (ppm): The basic algorithm, The ESCAPE SYMBOL, length of context, The Exclusion Principle, The Burrows-Wheeler Transform: Move-to-front coding, CALIC, JPEG-LS, Multi-resolution Approaches, Facsimile Encoding, Dynamic Markov Compression.

Unit – IV

Distortion criteria, Models, Scalar Quantization: The Quantization problem, Uniform Quantizer, Adaptive Quantization, Non uniform Quantization.

Unit-V

Advantages of Vector Quantization over Scalar Quantization, The Linde-Buzo-Gray Algorithm, Tree structured Vector Quantizers. Structured Vector Quantizers.

REFERENCES:

1. Khalid Sayood, Introduction to Data Compression, Morgan Kaufmann Publishers

2. Elements of Data Compression, Drozdek, Cengage Learning
3. Introduction to Data Compression, Second Edition, Khalid Sayood, The Morgan Kaufmann Series
4. Data Compression: The Complete Reference 4th Edition by David Salomon, Springer
5. Text Compression 1st Edition by Timothy C. Bell Prentice Hall

EMBEDDED SYSTEMS

Unit-I

Introduction to embedded systems: Classification, Characteristics and requirements, Applications

Unit-II

Timing and clocks in Embedded systems, Task Modeling and management, Real time operating system issues.

Unit-III

Signals, frequency spectrum and sampling, digitization (ADC, DAC), Signal Conditioning and Processing. Modeling and Characterization of Embedded Computation System.

Unit-IV

Embedded Control and Control Hierarchy, Communication strategies for embedded systems: Encoding and Flow control.

Unit-V

Fault-Tolerance, Formal Verification, Trends in Embedded Processor, OS, Development Language

References:

1. Prasad, Embedded /Real Time System, Concept, Design and Programming Black Book, Wiley India
2. R. Gupta, "Co-synthesis of Hardware and Software for Embedded Systems", Kluwer
3. Shibu K.V., "Introduction to Embedded Systems", TMH
4. Marwedel, "Embedded System Design", Springer

SEMANTIC WEB AND WEB SERVICES

UNIT I

Introduction to Semantic Web: History of Semantic Web, goals and vision, problems, Semantic Web Technologies, Layered Approach, Syntactic vs semantic web, Applications of semantic web.

UNIT II

Architecture: XML with Document Type Definitions and Schema, addressing and querying XML documents, RDF (Resource Description Framework), basic idea and syntax, querying in RQL, URI

UNIT III

Ontologies: Role of Ontology in intelligent information retrieval on web, OWL, Ontologies for different applications. Ontology engineering: constructing ontologies manually, reusing existing ontologies.

UNIT IV

Semantics: Kinds of semantics, use of semantics, Search Engines: Role of search Engines in intelligent retrieval of information on web, Semantic web browsers.

UNIT V

Logic and inference: examples of Monotonic rules: family relationships, monotonic rules: syntax and semantics, Non-monotonic rules: Motivation and syntax, Non-monotonic rule example: and Brokered Trade, Rule Mark-up XML: Monotonic and Non-Monotonic rules.

References:-

1. Salam, A. F., ed. Semantic Web Technologies and E-Business: Toward the Integrated Virtual Organization and Business Process Automation:. IGI Global, 2006.
2. Cardoso, Jorge, ed. Semantic Web Services: Theory, Tools and Applications: Theory, Tools and Applications. IGI Global, 2007.

3. Antoniou, Grigoris, and Frank Van Harmelen. A semantic web primer. MIT press, 2004.
4. Pascal Hitzler, Markus Krotzsch, Sebastian Rudolph, Foundations of Semantic Web Technologies, CRC Press
5. Daconta, Michael C., Leo J. Obrst, and Kevin T. Smith. The semantic web: a guide to the future of XML, web services, and knowledge management. John Wiley & Sons, 2003

INDUSTRIAL ECONOMICS AND PRINCIPLES OF MANAGEMENT

Industrial Economics:

Unit –1.Introduction: Nature and significance of Economics. Meaning of Science, Engineering and Technology and their relationship with economic development.

Unit –2. Basic Concept: The concept of demand and supply. Elasticity of Demand and Supply.

Indifference Curve Analysis, Price Effect, Income Effect and Substitution Effect.

Unit –3. Money and Banking: Functions of Money, Value of Money, Inflation and measures to control it. Brief idea of functions of banking system, viz., Commercial and central banking, Business fluctuations.

Management:

Unit –4. Introduction: Definition, Nature and Significance of Management,. Evaluation of Management thought, Contributions of Max Weber, Taylor and Fayol.

Unit –5. Human Behaviour: Factors of Individual Behaviour, Perception, Learning and Personality Development, Interpersonal Relationship and Group Behaviour.

References:

1. Dewett, K.K. / Modern Economic Theory/S.Chand & Co.
2. Luthers Fred/ Organizational Behaviour.
3. Prasad L.M./ Principles of Management.
4. A.W. Stonier & D.C. Horgne / A TextBook of Economic Theory/ Oxford Publishing House Pvt. Ltd.

COMPUTER GRAPHICS LAB

1. Implementation of line generation using slope's method, DDA and Bresenham's algorithms.
2. Implementation of circle generation using Mid-point method and Bresenham's algorithms.
3. Implementation of ellipse generation using Mid-point method.
4. Implementation of polygon filling using Flood-fill, Boundary –fill and scan line algorithms.
5. Implementation of 2-D transformation: Translation, Scaling, rotation, Mirror Reflection and shearing (write a menu driven program).
6. Implementation of line clipping using Cohen-Sutherland algorithm and Bisection Method.
7. Implementation of Polygon clipping using Sutherland-Hodgeman algorithms.
8. Implementation of 3-D geometric transformations: Translation, Scaling and rotation.
9. Implementation of curve generation using Interpolation methods.
10. Implementation of Curve generation using B-spline and Bezier curves.
11. Implementation of any one of back face removal algorithm (such that depth-buffer algorithm, Painter's algorithm, Warnock's algorithm, Scan line algorithm)

COMPILER DESIGN LAB

1. Simulation of Finite state Automata to recognize the tokens of various control statements.
2. Simulation of Finite state machine to distinguish among Integers, Real numbers with Exponents.
3. Program in LEX tool to recognize the tokens and return to token found for a C like language.
4. Parsing of arithmetical and algebraic expressions and equations.
5. Use of YACC tool to parse the statements of C like language.

Web Technology Lab

A Mini project based on different web technologies (eg. HTML, CSS, Java Script, JSP, JDBC & SQL).

OPERATING SYSTEMS

Unit -I

Introduction: Operating System and Function, Evolution of Operating System, Batch, Interactive, Time Sharing and Real Time System, System Protection. Operating System Structure: [04] , ' System Components, System Structure, Operating System Services.

Unit - II

Concurrent Processes: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Critical Section, Problem, Semaphores, Classical Problems in Concurrency, Inter Processes Communication, Process Generation, Process Scheduling.

Unit - III

CPU Scheduling: Scheduling Concept, Performance Criteria Scheduling Algorithm, Evolution, Multiprocessor Scheduling. Deadlock: [05] System Model, Deadlock Characterization, Prevention, Avoidance and Detection, Recovery From Deadlock Combined Approach.

Unit - IV

Memory Management: Basic Machine, Resident Monitor, Multiprogramming with Fixed Partition, Multiprogramming With Variable Partition, Multiple Base Register, Paging, Segmentation, Paged Segmentation, Virtual' Memory Concept, Demand Paging, Performance, Paged Replaced Algorithm, Allocation of Frames, Thrashing, Cache Memory Organization, Impact on Performance.

Unit - V

I/O Management & Disk Scheduling: I/O Devices and The Organization of I/O Function, I/O Buffering, Disk I/O, Operating System Design Issues. File System: File Concept, File Organization and Access Mechanism, File Directories, File Sharing, Implementation Issues.

References:

1. Milenekovie, "Operating System Concept", McGraw Hill.
2. Petersons, "Operating Systems", Addison Wesley.

3. Dietal, "An Introduction to Operating System", Addison Wesley.
4. Tannenbaum, "Operating System Design and Implementation", PHI.
5. Gary Nutt, "Operating System, A Modern Perspective", Addison Wesley.
6. Stalling, William, "Operating System", Maxwell Macmillan
7. Silveschatza, Peterson J, "Operating System Concepts", Willey.
8. Crowley, "Operating System", TMH.

COMPUTER NETWORKS

Unit -I

Introduction Concepts: Goals and Applications of Networks, Network structure and architecture, The OSI reference model, services, Network Topology Design - Delay Analysis, Back Bone Design, Local Access Network Design. Physical Layer Transmission Media, Switching methods, ISDN, Terminal Handling.

Unit-II

Medium Access sub layer: Medium Access sub layer - Channel Allocations, LAN protocols - ALOHA protocols - Overview of IEEE standards - FDDI. Data Link Layer - Elementary Data Link Protocols, Sliding Window protocols, Error Handling.

Unit - III

Network Layer: Network Layer - Point - to Pont Networks, routing, Congestion control Internetworking -TCP / IP - IP packet, IP address, IPv6. '

Unit - IV

Transport Layer: Transport Layer - Design issues, connection management, session Layer- Design issues, remote procedure call. Presentation Layer-Design issues, Data compression techniques, cryptography - TCP - Window Management.

Unit-V

Application Layer: Application Layer: File Transfer, Access and Management, Electronic mail, Virtual Terminals, Other application, Example Networks - Internet and Public Networks.

References:

1. Forouzen, "Data Communication and Networking", TMH
2. A.S. Tanenbaum, "Computer Networks", 3rd Edition, Prentice Hall India, 1997.
3. S. Keshav, "An Engineering Approach on Computer Networking", Addison Wesley, 1997

4. W. Stallings, "Data and Computer Communication", Macmillan Press, 1989.

ARTIFICIAL INTELLIGENCE

UNIT -I

Introduction

Introduction to Artificial Intelligence, Simulation of sophisticated & Intelligent Behavior in different area problem Solving in games, natural language, automated reasoning, visual perception, heuristic algorithm versus solution guaranteed algorithms.

UNIT - II

Understanding Natural Languages.

Parsing techniques, context free and transformational grammars, transition nets, augmented transition nets, Fillmore's grammars, Shanks Conceptual Dependency, grammar free analyzers, sentence generation, and translation.

UNIT III

Knowledge Representation

First order predicate calculus, Horn Clauses, Introduction to PROLOG, Semantic Nets, Partitioned Nets, Minsky frames, Case Grammar Theory, Production Rules Knowledge Base, The Interface System, Forward & Backward Deduction.

UNIT - IV

Expert System

Existing Systems (DENDRAL, MYCIN) domain exploration Meta Knowledge, Expertise Transfer, Self Explaining System

UNIT - V

Pattern Recognition

Introduction to Pattern Recognition, Structured Description, Symbolic Description, Machine perception, Line Finding, Interception Semantic & Model, Object Identification, Speech Recognition. Programming Language

Introduction to programming Language, LISP,
PROLOG References:

1. Charnick "Introduction to A.I.", Addison Wesley
2. Rich & Knight, "Artificial Intelligence"
3. Winston, "LISP", Addison Wesley
4. Marcellous, "Expert System Programming", PHI
5. Elamie, "Artificial Intelligence", Academic Press
6. Lioyed, "Foundation of Logic Processing", Springer Verlag

Graph Theory

Unit-I

Graphs, Sub graphs, some basic properties, various example of graphs & their sub graphs, walks, path & circuits, connected graphs, disconnected graphs and component, euler graphs, various operation on graphs, Hamiltonian paths and circuits, the traveling sales man problem.

Unit-II

Trees and fundamental circuits, distance diameters, radius and pendent vertices, rooted and binary trees, on counting trees, spanning trees, fundamental circuits, finding all spanning trees of a graph and a weighted graph, algorithms of primes, Kruskal and Dijkstra Algorithms.

Unit-III

Cuts sets and cut vertices, some properties, all cut sets in a graph, fundamental circuits and cut sets, connectivity and reparability, network flows Planer graphs, combinatorial and geometric dual: Kuratowski graphs, detection of planarity, geometric dual, Discussion on criterion of planarity, thickness and crossings.

Unit-IV

Vector space of a graph and vectors, basis vector, cut set vector, circuit vector, circuit and cut set subspaces, Matrix representation of graph - Basic concepts; Incidence matrix, Circuit matrix, Path matrix, Cut-set matrix and Adjacency matrix. Coloring, covering and partitioning of a graph, chromatic number, chromatic partitioning, chromatic polynomials, matching, covering, four color problem Discussion of Graph theoretic algorithm wherever required.

References:

1. Deo, N, Graph theory with applications to Engineering and Computer Science, PHI
2. Gary Chartrand and Ping Zhang, Introduction to Graph Theory, TMH
3. Robin J. Wilson, Introduction to Graph Theory, Pearson Education
4. Harary, F, Graph Theory, Narosa
5. Bondy and Murthy: Graph theory and application. Addison Wesley.
6. V. Balakrishnan, Schaum's Outline of Graph Theory, TMH
7. Geir Agnarsson, Graph Theory: Modeling. Applications and Algorithms, Pearson Education

Operations Research

Unit-I

Linear programming problems: Linear programming problems (LPP)- Formulation of a LPP- graphical method-Simplex method- Revised simplex method-two phase method-Dual simplex method-Primal-Dual Problem (Emphasis should be on algorithms and problems).

Unit-II

Transportation and Assignment problem: Principles of duality-Interpretation sensitivity analysis-degeneracy-Integer (Linear) programming, branch and bound method computational procedure application of IP,0-1 linear programming problem, Knapsack problem, facility location problem assignment problems, mathematical formulation, fundamental theorem, Hungarian method for solving an assignment problem , variation of an assignment problem, Application(Emphasis should be more on problem than theory).

Unit-III

CPM and PERT: CPM and PERT- network diagram-Events and activities- project planning-reducing critical events and activities-critical path calculations-Examples- Resources and man power leveling. Sequencing problems-travelling salesman problems –machine-scheduling problem (Job-shop).

Unit-IV

Replacement problems and Inventory models: Replacement problems-capital equipment-Discounting costs-Replacement in anticipation of failure-Group replacement –stochastic nature underlying the failure phenomenon. Inventory models-various costs-Deterministic inventory models-Economic lot sizes –Price breaks –Finite storage.

Unit-V

Inventory Model Application and Dynamic programming: Single period inventory model with shortest cost-stochastic models-Application of inventory models. Dynamic programming formulation-Investment problem-general allocation problem –storage coach problem-production scheduling.

References:

1. H.A. TAHA, "Operations Research- An Introduction", Macmillan 1976.
2. Hillier and Lieberman: "Introduction to operation research" , (1990) Mc Graw Hill, Company.
3. Ecker and Kuperfersch mid: "Introduction to Operation research" (1988), John Wiley & Sons.
4. B.E. Gillet, "introduction to Operation Research-A Computer oriented Algorithmic Approach", McGraw Hill 1989
5. K.Swarup, P. K. Gupta & A. Manmohan, "Operation Research", S.chand 1978.

QUALITY MANAGEMENT

Unit-1

Quality Concepts: Evolution of Quality Control, concept change, TQM Modern concept, Quality concept in design, Review of design, Evolution of proto type. Control on Purchased Product Procurement of various products, evaluation of supplies, capacity verification, Development of sources, procurement procedure. Manufacturing Quality -Methods and techniques for manufacture, inspection and control of product, quality insales and services, guarantee, analysis of claims.

UNIT-II

Quality Management Organization structure and design, quality function, decentralization, designing and fitting, organization for different type products and company, economics of quality value and contribution, quality cost, optimizing quality cost, seduction program. Human Factor in quality Attitude of top management, cooperation of groups, operators attitude, responsibility, causes of apparatus error and corrective methods.

UNIT-III Control

Charts, Theory of control charts, measurement range, construction and analysis of R charts, process capability study, use of control charts. Attributes of Control Chart Defects, construction and analysis of charts, improvement by control chart, variable sample size, construction and analysis of C charts.

UNIT-IV

Defects diagnosis and prevention defect study, identification and analysis of defects, correcting measure, factors affecting reliability, MTTF, calculation of reliability, building reliability in the product, evaluation of reliability, interpretation of test results, reliability control, maintainability, zero defects, quality circle.

UNIT-V

ISO-9000 and its concept of Quality Management, ISO 9000 series, Taguchi method, JIT in some details.

Text/Reference Books:

Gen. H. Lal, "Total Quality Management", Eastern Limited, 1990.

Greg Bounds, "Beyond Total Quality Management", McGraw Hill, 1994.

Menon, H.G, "TQM in New Product manufacturing", McGraw Hill 1992.

PRODUCT DEVELOPMENT

UNIT-I

Concept of Product, definition and scope. Design definitions, old and new design methods, design by evolution, examples such as evolution of sewing M/C, bicycle, safety razor etc., need based developments, technology based developments physical reliability & economic feasibility of design concepts.

UNIT-II

Morphology of design, divergent, transformation and convergent phases of product design, identification of need, Analysis of need. Design criteria; functional, aesthetics, ergonomics, form, shape, size, colour. Mental blocks, Removal blocks, Ideation techniques, Creativity, Check list.

UNIT-III

Transformations, Brainstorming & Syntetics, Morphological techniques. Utility Concept, Utility Value, Utility Index, Decision making under Multiple Criteria. Economic aspects, Fixed and variable costs, Break-even analysis.

UNIT-IV

Reliability considerations, Bath tub curve, Reliability of systems in series and parallel, Failure rate, MTTF and MTBF, Optimum spares from Reliability considerations. Design of display and controls, Man-machine interface, Compatibility of displays and controls. Ergonomic aspects, Anthropometric data and its importance in design. Application of Computers in Product development & design.

UNIT-V

Existing techniques, such as work-study, SQC etc. for improving method & quality of product. Innovation versus Invention. Technological Forecasting. Use of Standards for Design.

Text/Reference Books:

1. A.K. Chitambar & R.C. Gupta "Product design & Manufacturing" - Prentice Hall (EE)
2. R.P. Crawford, "The Technology of creative Thinking" Prentice Hall.
3. C.D. Cain, "Product Design & Decision" Business Books.

4. C.D. Cain, "Engg. Product Design" Bussiness Books.

ORGANIZATIONAL BEHAVIOR

Introduction to Organizations and individuals:

What is organization, components of organization, nature and variety of organizations (in terms of objective structure etc.), Models of analyzing organizational phenomena, organizational and business variables, organizations in the Indian context, institutions and structures, basic role in the organization, etc. perceptions, attitudes, motives (achievement, power and affiliation), commitment, values, creativity and other personality factors, profile of manager and a entrepreneur.

Interpersonal and group processes:

Interpersonal trust under standing the other person from his/her point of view, Interpersonal communication, listening feedback, counseling, transactional analysis, self fulfilling prophecy, etc. leadership, motivating people, working as a member of a team, team functioning, team decision making, team conflict resolution, team problem solving.

Organizational structure and integrating Interpersonal and group dynamic elements of structure, functions of structures, determinants of structures, dysfunctional ties of structures, structure – technology? Environment people relationships, principle-underlying design of organizational politics, issues of power and authority, organizational communications, organizational change, integrating cases(s), case method and lectures should be supplemented with a variety of other methodologies such as feedback on questionnaires and tests, role plays, and behavior simulation exercise.

References :

1. Jit S Chandan "Organizational Behavior", Vikas
2. M.N. Mishra : "Organization Behavior", Vikas
3. Arnold, John, Robertson, Ivan I. and Cooper, Cary, I., " Work Psychology: understanding human behavior in the workplace", Macmillan India Ltd., Delhi. 1996.
4. Dwivedi, RS., Human relations and organizational behavior: a global perspective, Macmillan India Ltd., Delhi, .1995.
5. Hersey and Blanchard (6th ed.). "Management of organizational behavior L utilising human resources", Prentice Hall of India Pvt. Ltd., New Delhi, 1996.

6. Robbins (4th ed.), "Essentials of organizational behavior", Prentice Hall of India Pvt. Ltd., New Delhi, 1995.
7. Luthans Fred., "Organizational Behavior", McGraw Hill, 1998.

OPERATING SYSTEMS LAB

1. Simulation of the scheduling algorithms a) Round Robin b) SJF c) FCFS d) Priority.
2. Simulation of MUTEX and SEMAPHORES.
3. Simulation of Banker's deadlock Avoidance and Prevention Algorithms.
4. Implementation of Process Synchronizations (Reader's-Writer's, Sleeping Barber, and Dining Philosopher's Problem).
5. Simulation of Page replacement algorithms a) FIFO b) LRU c) LFU
6. Simulation of Paging Techniques of memory management.
7. Simulation of file allocation strategies a) Sequential b) Index c) Linked
8. Simulation of organization techniques a) Single level Directory b) Two level c) Hierarchical d)

DAG

Computer Network Lab

1. Implementation of sender and receiver for testing of connection between two.
2. Implementation of one program on socket programming
3. Implementation of Simplex Protocol with noisy channel
4. Implementation of sliding window protocol
5. Implementation of any one protocol of application layer
6. Implementation of any one protocol of network layer
7. Implementation of RSA Algorithm for encryption and decryption
8. Implementation of DSA Algorithm for encryption and decryption

Artificial Intelligent Lab

1. Write a LISP program to solve water-jag problem using heuristic function.
2. Create a compound object using Prolog.
3. Write a Prolog program to show advantage and disadvantage of green and red cuts.
4. Write a Prolog program to use BEST FIRST SEARCH applied to the 8-puzzle problem.
5. Implement SLEEPEST ASCENT HILL CLIMBING using LISP.
6. Implement COUNT PROPAGATION NETWORK using Prolog.
7. Implement a) Forward chaining b) Backward chaining c) Problem Reduction.

ADVANCE COMPUTER ARCHITECTURE

Unit – I Introduction:

Introduction to parallel computing, need for parallel computing, parallel architectural classification schemes, Flynn's, Fengs classification, performance of parallel processors, distributed processing, processor and memory hierarchy, bus, cache & shared memory, introduction to super scalar architectures, quantitative evaluation of performance gain using memory, cache miss/hits.

Unit – II Multi-core architectures:

Introduction to multi-core architectures, issues involved into writing code for multi-core architectures, development of programs for these architectures, program optimizations techniques, building of some of these techniques in compilers, OpenMP and other message passing libraries, threads, mutex etc.

Unit – III Multi-threaded architectures:

Parallel computers, Instruction level parallelism (ILP) vs. thread level parallelism (TLP), Performance issues: Brief introduction to cache hierarchy and communication latency, Shared memory multiprocessors, General architectures and the problem of cache coherence, Synchronization primitives: Atomic primitives; locks: TTS, ticket, array; barriers: central and tree; performance implications in shared memory programs; Chip multiprocessors: Why CMP (Moore's law, wire delay); shared L2 vs. tiled CMP; core complexity; power/performance; Snoopy coherence: invalidate vs. update, MSI, MESI, MOESI, MOSI; performance trade-offs; pipelined snoopy bus design; Memory consistency models: SC, PC, TSO, PSO, WO/WC, RC; Chip multiprocessor case studies: Intel Montecito and dual-core, Pentium4, IBM Power4, Sun Niagara

Unit – IV Compiler optimization issues:

Introduction to optimization, overview of parallelization; Shared memory programming, introduction to OpenMP; Dataflow analysis, pointer analysis, alias analysis; Data dependence analysis, solving data dependence equations (integer linear programming problem); Loop optimizations; Memory hierarchy issues in code optimization.

Unit –V Operating system issues and applications:

Operating System issues for multiprocessing Need for pre-emptive OS; Scheduling Techniques, Usual OS scheduling techniques, Threads, Distributed scheduler, Multiprocessor scheduling, Gang scheduling; Communication between processes, Message boxes, Shared memory; Sharing issues and Synchronization, Sharing memory and other structures, Sharing I/O devices, Distributed

Semaphores, monitors, spin-locks, Implementation techniques on multi-cores; OpenMP, MPI and case studies Case studies from Applications: Digital Signal Processing, Image processing, Speech processing.

Books :

1. Hwang, “ Advanced Computer Architecture”, New Age International
2. Quin, “ Parallel Computing, Theory & Practices”, McGraw Hill

Cryptography and Network Security

Unit-1

Introduction to security attacks, services and mechanisms, Introduction to cryptography.

Conventional Encryption: Conventional to cryptography model, classical encryption techniques- substitution ciphers and transposition ciphers, cryptanalysis, stereography, stream and block cipher. Modern block ciphers: Block ciphers principles, Shannon's theory of confusion and diffusion, feistel structure, Data encryption standards(DES), strength of DES, Differential and linear cryptanalysis of DES, Block cipher modes of operation, Triple DES, IDEA encryption and Decryption, strength of IDEA, Confidentiality using conventional encryption, traffic confidentiality, key distribution, random number generation.

Unit-2

Introduction to group, ring and field, prime and relative prime numbers, modular arithmetic, Fermat's & Euler's theorem, primality testing, euclid's algorithm, Chinese remainder theorem, discrete logarithms. Principle of public key cryptosystems, RSA algorithm, security of RSA, key management, Diffie Hellman key exchange algorithms, Introductory idea of Elliptic curve cryptography, ElGamal Encryption.

Unit-3

Message Authentication & Hash functions: Authentication requirements, Authentication functions, message Authentication codes, Hash functions, Birthday attacks, Security of hash function & MACS, MD5 message digest algorithms, Secure hash algorithm(SHA).

Digital signatures: Digital signatures, Authentication protocol, Digital signatures Standards(DSS), proof of Digital signatures algorithm.

Unit-4

Authentication Applications: Kerberos & X.509, Directory authentication service, Electronic Mail Security Pretty Good Privacy(PGP), S/MIME.

Unit-5

IP Security: Architecture, Authentication Header, Encapsulation security payloads, Combining security associations, Key management.

Web Security: Secure socket layer & Transport Layer Security, Secure Electronic Transaction (SET).

System Security: Intruders, Viruses & related threats, Firewall design principles, Trusted systems.

Reference:

- 1 William Stalling, "Cryptography and Network Security: Principles and Practice", Prentice Hall, New Jersey.
- 2 Johannes A. Buchman, "Introduction to Cryptography", SpringerVerlag.

HIGH SPEED NETWORKS

UNIT I

Frame Relay Networks – Asynchronous transfer mode–ATM Protocol Architecture, ATM logical Connection – ATM Cell – ATM Service Categories – AAL. High Speed LAN's: Fast Ethernet – Gigabit Ethernet– Fiber Channel – Wireless LAN's, WiFi and WiMax Networks applications, requirements – Architecture of 802.11.

UNIT II

Queuing Analysis – Queuing Models – Single Server Queues – Effects of Congestion – Congestion Control – Traffic Management – Congestion Control in Packet Switching Networks – Frame Relay Congestion Control.

UNIT III

TCP Flow control – TCP Congestion Control – Retransmission – Timer Management – Exponential RTO backoff – KARN's Algorithm – Window management – Performance of TCP over ATM. Traffic and Congestion control in ATM – Requirements – Attributes – Traffic Management Frame work, Traffic Control – ABR traffic Management – ABR rate control, RM cell formats – ABR Capacity allocations – GFR traffic management.

UNIT IV

Integrated Services Architecture – Approach, Components, Services- Queuing Discipline– FQ – PS – BRFQ – GPS – WFQ – Random Early Detection – Differentiated Services.

UNIT V

RSVP – Goals & Characteristics, Data Flow, RSVP operations – Protocol Mechanisms– Multiprotocol Label Switching – Operations, Label Stacking – Protocol details – RTP– Protocol Architecture – Data

REFERENCES:

1. William Stallings, "High speed networks and internet", Second Edition, Pearson Education, 2002
2. Warland, Pravin Varaiya, "High performance communication networks", Second Edition , Jean Harcourt Asia Pvt. Ltd., , 2001
3. Irvan Pepelnjk, Jim Guichard, Jeff Apcar, "MPLS and VPN architecture", Cisco Press, Volume 1 and 2, 2003.
4. Abhijit S. Pandya, Ercan Sea, "ATM Technology for Broad Band Telecommunication Networks", CRC Press, New York, 2004

ANDROID OPERATING SYSTEM

UNIT I

Android OS Android Software Stack, Activities and Applications, Activity Life Cycles, Activity Stacks, Activity States, Resources, Android OS vs. IOS

UNIT II

User Interfaces Views, Layouts, Android Widgets, UI XML Specifications, Explicit Intents, Implicit Intents, Event Broadcasting with Intents, Event Reception with Broadcast Receivers, Adapters and Data Binding.

UNIT III

Multimedia Audio, Video, Camera, Playing Audio and Video, Recording Audio and Video, Using the Camera to Take and Process Pictures

UNIT IV

Networking Internet Access, HTML and XML Parsing, Wi-Fi

UNIT V

Touchscreen Capturing Touch Events, Touchscreen Gesture Recognition

REFERENCES:

1. Rito Meier. "Professional Android 2 Application Development." Wiley Publishing, Inc.
2. SayedHashimi, SatyaKomatineni, Dave MacLean. "Pro Android 2." APRESS.
3. Mark Murphy. "Beginning Android 2." APRESS.
4. Carmen Delessio, Lauren Darcey "Android Application Development" Pearson
5. J.F. DiMarzio "Android a programming guide" TMH

Service Oriented Architecture

Unit-I

Introduction: SOA and MSA Basics: Service Orientation in Daily Life, Evolution of SOA and MSA. Service oriented Architecture and Microservices architecture – Drivers for SOA, Dimensions of SOA, Conceptual Model of SOA, Standards and Guidelines for SOA, Emergence of MSA.

Enterprise-Wide SOA: Considerations for Enterprise-wide SOA, Strawman Architecture for

Enterprise-wide SOA, Enterprise SOA Reference Architecture, Object-oriented Analysis and Design (OOAD) Process, Service-oriented Analysis and Design (SOAD) Process, SOA Methodology for Enterprise

Unit-II

Service-Oriented Applications: Considerations for Service-oriented Applications, Patterns for SOA, Pattern-based Architecture for Service-oriented Applications, Composite Applications, Composite Application Programming Model.

Service-Oriented Analysis and Design: Need for Models, Principles of Service Design, Nonfunctional Properties for Services, Design of Activity Services (or Business Services), Design of Data Services, Design of Client Services, Design of Business Process Services.

Unit-III

Technologies for SOA: Technologies for Service Enablement, Technologies for Service Integration, Technologies for Service Orchestration.

SOA Governance and Implementation: Strategic Architecture Governance, Service Design-time Governance, Service Run-time Governance, Approach for Enterprise- wide SOA Implementation.

Unit-IV

Big Data and SOA: Concepts, Big Data and its characteristics, Technologies for Big Data, Service-orientation for Big Data Solutions.

Business Case for SOA: Stakeholder Objectives, Benefits of SOA, Cost Savings, Return on Investment (ROI), Build a Case for SOA

Unit-V

SOA Best Practices: SOA Strategy – Best Practices, SOA Development – Best Practices, SOA Governance – Best Practices.

EA and SOA for Business and IT Alignment: Enterprise Architecture, Need for Business and It Alignment, EA and SOA for Business and It Alignment

Text books:

1. Shankar Kambhampaty; Service - Oriented Architecture & Microservices Architecture: For Enterprise, Cloud, Big Data and Mobile; Wiley; 3rd Edition; 2018; ISBN: 9788126564064.
2. Icon Group International; The 2018-2023 World Outlook for Service-Oriented Architecture (SOA) Software and Services; ICON Group International; 1st Edition, 2017; ASIN: B06WGPN8YD.
3. Thomas Erl; Service Oriented Architecture Concepts Technology & Design; Pearson Education Limited; 2015; ISBN-13: 9788131714904.
4. Guido Schmutz, Peter Welkenbach, Daniel Liebhart; Service Oriented Architecture An Integration Blueprint; Shroff Publishers & Distributors; 2010; ISBN-13: 9789350231081

DATA MINING & DATA WAREHOUSING

Unit-I: Introduction

Review of basic concepts of data warehousing and data mining, reasons for their use, benefits and problems arising.

Unit-II: Data Warehouse Design

Data warehouse logical design: star schemas, fact tables, dimensions, other schemas, materialized, views, Data warehouse physical design: hardware and i/o considerations, parallelism, indexes.

Unit-III: Data Warehouse Implementation

Data warehousing technologies and implementations: data extraction, transportation, transformation, loading and refreshing. Data warehouse support in SQL Server and Oracle.

Unit- IV: Data Mining

From data warehousing to data mining: OLAP architectures, design and query processing. SQL, extensions for OLAP.

Unit-V: Data Mining Approaches

Data mining approaches and methods: concept description, classification, association rules, clustering, Mining complex types of data, Research trends in data warehousing and data mining.

Books:

- 1 Data Mining - Concepts and Techniques by Jiawei Han and Micheline Kamber, Morgan Kaufmann 2006.
- 2 Data Mining: Practical Machine Learning Tools and Techniques with Java Implementations by Ian H. Witten and Eibe Frank, Morgan Kaufmann 2000.
- 3 Data Mining: Introductory and Advanced Topics by Margaret Dunham, Prentice Hall 2003.
- 4 Data Mining with Microsoft SQL Server 2000 Technical Reference Microsoft Press.

Distributed Databases

Unit-I: Introduction

Architecture of distributed systems: A detailed review of distributed system architecture (network operating system, distributed operating systems, etc.) will be presented leading to distributed database systems. This will then be categorized into (a) federated database systems, (b) multi-database systems, and (c) Client/Server systems.

Advanced transaction model: For managing data processing on distributed platform the conventional transaction model needs some improvements. Discussion of some advanced transaction models suitable for different types of distributed database systems.

Unit-II: Workflow

It is a unit of business processing. From conventional viewpoint it is a set of tightly linked atomic processing units which requires special concurrency control and commit protocols. Discussion of existing ways of handling workflows.

Unit-III: Query processing and Optimization: On distributed systems a query may be fragmented for processing on multiple nodes. This give rise to the problem of query fragmentation and distribution which must be addressed for improving performance.

Unit-IV: Application distribution: To support parallel and concurrent processing of transactions processing application have to be distributed. This gives rise to application recovery problem. This course will explore new ways of managing application recovery which is more complex than database recovery.

Unit-V: Transaction management, commit protocol and database recovery: These are system related issues. We will discuss commonly used schemes and advanced protocols for managing these activities.

Buffer management: Database maintains their own buffer for processing transactions.

Books:

1. Distributed Systems: Concept and Design. Coulouris, Dollimore, and Kindberg. AW.
2. Distributed Database Principles and Systems. Ceri and Pelagatti. McGraw Hill.
3. Recovery Mechanisms in Database Systems. Kumar and Hsu, Prentice Hall.
4. Concurrency Control and Recovery in Database Systems. Bernstein, Hadzilacos and Goodman, AW.

Bioinformatics

Unit I: Introduction

Bioinformatics objectives and overviews, Interdisciplinary nature of Bioinformatics, Data integration, Data analysis, Major Bioinformatics databases and tools. Metadata: Summary & reference systems, finding new type of data online.

Molecular Biology and Bioinformatics: Systems approach in biology, Central dogma of molecular biology, problems in molecular approach and the bioinformatics approach, Overview of the bioinformatics applications.

Unit II: The Information Molecules and Information Flow

Basic chemistry of nucleic acids, Structure of DNA, Structure of RNA, DNA Replication, - Transcription, - Translation, Genes- the functional elements in DNA, Analyzing DNA, DNA sequencing. Proteins: Amino acids, Protein structure, Secondary, Tertiary and Quaternary structure, Protein folding and function, Nucleic acid-Protein interaction.

Unit III: Perl

Perl Basics, Perl applications for bioinformatics- Bioperl, Linux Operating System, Understanding and Using Biological Databases, Java clients, CORBA, Introduction to biostatistics.

Unit IV: Nucleotide sequence data

Genome, Genomic sequencing, expressed sequence tags, gene expression, transcription factor binding sites and single nucleotide polymorphism. Computational representations of molecular biological data storage techniques: databases (flat, relational and object oriented), and controlled vocabularies, general data retrieval techniques: indices, Boolean search, fuzzy search and neighboring, application to biological data warehouses.

Unit V: Biological data types and their special requirements: sequences, macromolecular structures, chemical compounds, generic variability and its connection to clinical data. Representation of patterns and relationships: alignments, regular expressions, hierarchies and graphical models.

Books:

1. O'Reilly, "Developing Bioinformatics computer skills", Indian Edition's publication

2. Rastogi, Mendiratta, Rastogi, "Bioinformatics concepts, skills & Applications", CBS Publishers
3. Rashidi, Hooman and Lukas K. Buehler, "Bioinformatics Basic Applications" CRC Press.
4. "Bioinformatics" , Addison Wesley
5. Stephen Misner & Stephen Krawetz, " Bioinformatics- Methods & Protocols"

Quantum Computing

Unit-I

Fundamental Concepts: Global Perspectives, Quantum Bits, Quantum Computation, Quantum Algorithms, Quantum Information, Postulates of Quantum Mechanisms.

Unit-II

Quantum Computation: Quantum Circuits – Quantum algorithms, Single Orbit operations, Control Operations, Measurement, Universal Quantum Gates, Simulation of Quantum Systems, Quantum Fourier transform, Phase estimation, Applications, Quantum search algorithms – Quantum counting – Speeding up the solution of NP – complete problems – Quantum Search for an unstructured database.

Unit-III

Quantum Computers: Guiding Principles, Conditions for Quantum Computation, Harmonic Oscillator Quantum Computer, Optical Photon Quantum Computer – Optical cavity Quantum electrodynamics, Ion traps, Nuclear Magnetic resonance

Unit-IV

Quantum Information: Quantum noise and Quantum Operations – Classical Noise and Markov Processes, Quantum Operations, Examples of Quantum noise and Quantum Operations – Applications of Quantum operations, Limitations of the Quantum operations formalism, Distance Measures for Quantum information.

Unit-V

Quantum Error Correction: Introduction, Shor code, Theory of Quantum Error – Correction, Constructing Quantum Codes, Stabilizer codes, Fault – Tolerant Quantum Computation, Entropy and information – Shannon Entropy, Basic properties of Entropy, Von Neumann, Strong Sub Additivity, Data Compression, Entanglement as a physical resource .

Text books:

1. Micheal A. Nielsen. & Issac L. Chiang, “Quantum Computation and Quantum Information”, Cambridge University Press, Fint South Asian edition, 2002.
2. Eleanor G. Rieffel, Wolfgang H. Polak , “Quantum Computing - A Gentle Introduction” (Scientific and Engineering Computation) Paperback – Import, Oct 2014
3. Computing since Democritus by Scott Aaronson, Computer Science: An Introduction by N. David Mermin 5. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists.

MOBILE COMPUTING

Unit – I

Introduction, issues in mobile computing, overview of wireless telephony: cellular concept, GSM: air-interface, channel structure, location management: HLR-VLR, hierarchical, handoffs, channel allocation in cellular systems, CDMA, GPRS.

Unit - II

Wireless Networking, Wireless LAN Overview: MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, data broadcasting, Mobile IP, WAP: Architecture, protocol stack, application environment, applications.

Unit – III

Data management issues, data replication for mobile computers, adaptive clustering for mobile wireless networks, File system, Disconnected operations.

Unit - IV

Mobile Agents computing, security and fault tolerance, transaction processing in mobile computing environment.

Unit – V

Ad Hoc networks, localization, MAC issues, Routing protocols, global state routing (GSR), Destination sequenced distance vector routing (DSDV), Dynamic source routing (DSR), Ad Hoc on demand distance vector routing (AODV), Temporary ordered routing algorithm (TORA), QoS in Ad Hoc Networks, applications.

Books:

1. J. Schiller, Mobile Communications, Addison Wesley.
2. A. Mehrotra , GSM System Engineering.
3. M. V. D. Heijden, M. Taylor, Understanding WAP, Artech House.
4. Charles Perkins, Mobile IP, Addison Wesley.
5. Charles Perkins, Ad hoc Networks, Addison Wesley.

Neural Networks

Unit-I: Neurocomputing and Neuroscience

Historical notes, human Brain, neuron Model, Knowledge representation, AI and NN. Learning process: Supervised and unsupervised learning, Error correction learning, competitive learning, adaptation, and statistical nature of the learning process.

Unit-II: Data processing

Scaling, normalization, Transformation (FT/FFT), principal component analysis, regression, covariance matrix, eigen values & eigen vectors. Basic Models of Artificial neurons, activation Functions, aggregation function, single neuron computation, multilayer perceptron, least mean square algorithm, gradient descent rule, nonlinearly separable problems and bench mark problems in NN.

Unit-III

Multilayered network architecture, back propagation algorithm, heuristics for making BP-algorithm performs better. Accelerated learning BP (like recursive least square, quick prop, RPROP algorithm), approximation properties of RBF networks and comparison with multilayer perceptron.

Unit-IV

Recurrent network and temporal feed-forward network, implementation with BP, self organizing map and SOM algorithm, properties of feature map and computer simulation. Principal component and Independent component analysis, application to image and signal processing.

Unit-V

Complex valued NN and complex valued BP, analyticity of activation function, application in 2D information processing. Complexity analysis of network models. Soft computing. Neuro-Fuzzy-genetic algorithm Intergration.

REFERENCES:

1. I.J.A. Anderson, An Introduction to Neural Networks, MIT

2. Hagen Demuth Beale, Neural Network Design, Cengage Learning

3. Laurene V. Fausett, "Fundamentals of Neural Networks : Architectures, Algorithms and Applications", Pearson

India

4. Munesh Chandra Trivedi, NN Jani, Artificial Neural Network Technology, Khanna Publishing House

5. Kosko, Neural Network and Fuzzy Sets, PHI 5. Hagan, Neural Network Design w/CD, Cengage Learning

NATURAL LANGUAGE PROCESSING

Unit-I

Introduction to Natural Language Understanding: The study of Language, Applications of NLP, Evaluating Language Understanding Systems, Different levels of Language Analysis, Representations and Understanding, Organization of Natural language Understanding Systems, Linguistic Background: An outline of English syntax.

Unit-II

Introduction to semantics and knowledge representation, Some applications like machine translation, database interface.

Unit-III

Grammars and Parsing: Grammars and sentence Structure, Top-Down and Bottom-Up Parsers, Transition Network Grammars, Top-Down Chart Parsing. Feature Systems and Augmented Grammars: Basic Feature system for English, Morphological Analysis and the Lexicon, Parsing with Features, Augmented Transition Networks.

Unit-IV

Grammars for Natural Language: Auxiliary Verbs and Verb Phrases, Movement Phenomenon in Language, Handling questions in Context-Free Grammars. Human preferences in Parsing, Encoding uncertainty, Deterministic Parser.

Unit-V

Ambiguity Resolution: Statistical Methods, Probabilistic Language Processing, Estimating Probabilities, Part-of-Speech tagging, Obtaining Lexical Probabilities, Probabilistic Context-Free Grammars, Best First Parsing. Semantics and Logical Form, Word senses and Ambiguity, Encoding Ambiguity in Logical Form.

Books:

- 1 Akshar Bharti, Vineet Chaitanya and Rajeev Sangal, *NLP: A Paninian Perspective*,

- 2 James Allen, *Natural Language Understanding*, 2/e, Pearson Education, 2003
- 3 D. Jurafsky, J. H. Martin, *Speech and Language Processing*, Pearson Education, 2002
- 4L.M. Iivansca, S. C. Shapiro, *Natural Language Processing and Language Representation*
- 5 T. Winograd, *Language as a Cognitive Process*, Addison-Wesley

FUZZY SYSTEMS

UNIT 1

Introduction, Basic Types, Basic Concepts, Representations of Fuzzy Sets, Extension Principle for Fuzzy Sets, Types of Operations. Fuzzy Complements, Fuzzy Intersections: t - Norms., Fuzzy Unions: t -Conorms, Combinations of Operations. Aggregation Operations. Arithmetic: Fuzzy Numbers, Linguistic Variables, Arithmetic Operations on Intervals, Arithmetic Operations on Fuzzy Numbers, Fuzzy Equations

UNIT 2

Crisp versus Fuzzy Relations, Projections and Cylindric Extensions, Binary Fuzzy Relations, Binary Relations on a Single Set. Fuzzy Equivalence Relations, Fuzzy Compatibility Relations. Fuzzy Ordering Relations, Fuzzy Morphisms, $\sup$ - i Compositions of Fuzzy Relations., $\inf$ - Compositions of Fuzzy Relations.

UNIT 3

Fuzzy Measures, Fuzzy Sets and Possibility Theory, Classical Logic: An Overview. Multivalued Logics. Fuzzy Propositions. Fuzzy Quantifiers. Linguistic Hedges. Inference from Conditional Fuzzy Propositions. Inference from Conditional and Qualified Propositions. Inference from Quantified Propositions, Information and Uncertainty, Nonspecificity of Fuzzy Sets. Fuzziness of Fuzzy Sets. Principles of Uncertainty

UNIT 4

Fuzzy Expert Systems: An Overview. Fuzzy Implications. Selection of Fuzzy Implications. Multiconditional Approximate Reasoning. The Role of Fuzzy Relation Equations, Fuzzy Controllers: Overview, Fuzzy Neural Networks. Fuzzy Automata. Fuzzy Dynamic Systems.

UNIT 5

Fuzzy Databases. Fuzzy Information Retrieval, Individual Decision Making, Multiperson Decision Making, Multicriteria Decision Making, Multistage Decision Making, Fuzzy Systems and Genetic Algorithms.

Reference:

- George J. Klir, Bo Yuan, "Fuzzy Sets and Fuzzy Logic", PHI
- Witold Pedrycz and Fernando Gomide. "An Introduction to Fuzzy Sets", PHI

SOFTWARE PROJECT MANAGEMENT

UNIT-I: Introduction and Software Project Planning

Fundamentals of Software Project Management (SPM), Need Identification, Vision and Scope document, Project Management Cycle, SPM Objectives, Management Spectrum, SPM Framework, Software Project Planning, Planning Objectives, Project Plan, Types of project plan, Structure of a Software Project Management Plan, Software project estimation, Estimation methods, Estimation models, Decision process.

UNIT-II: Project Organization and Scheduling

Project Elements, Work Breakdown Structure (WBS), Types of WBS, Functions, Activities and Tasks, Project Life Cycle and Product Life Cycle, Ways to Organize Personnel, Project schedule, Scheduling Objectives, Building the project schedule, Scheduling terminology and techniques, Network Diagrams: PERT, CPM, Bar Charts: Milestone Charts, Gantt Charts.

UNIT-III: Project Monitoring and Control

Dimensions of Project Monitoring & Control, Earned Value Analysis, Earned Value Indicators: Budgeted Cost for Work Scheduled (BCWS), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Interpretation of Earned Value Indicators, Error Tracking, Software Reviews, Types of Review: Inspections, Deskchecks, Walkthroughs, Code Reviews, Pair Programming.

UNIT-IV: Software Quality Assurance and Testing

Testing Objectives, Testing Principles, Test Plans, Test Cases, Types of Testing, Levels of Testing, Test Strategies, Program Correctness, Program Verification & validation, Testing Automation & Testing Tools, Concept of Software Quality, Software Quality Attributes, Software Quality Metrics and Indicators, The SEI Capability Maturity Model CMM), SQA Activities, Formal SQA Approaches: Proof of correctness, Statistical quality assurance, Cleanroom process.

UNIT-V: Project Management and Project Management Tools

Software Configuration Management: Software Configuration Items and tasks, Baselines, Plan for Change, Change Control, Change Requests Management, Version Control, Risk Management: Risks and risk types, Risk Breakdown Structure (RBS), Risk Management Process: Risk identification, Risk analysis, Risk planning, Risk monitoring, Cost Benefit Analysis, Software Project Management Tools: CASE Tools, Planning and Scheduling Tools, MS-Project.

Books:

- 1) Software Project Management by M. Cotterell
- 2) Information Technology Project Management
- 3) Management Information and Control by
- 4) Software Project Managemnet by S. A. Kelkar

SOFTWARE QUALITY ENGINEERING

UNIT-I:

Introduction 8 Defining Software Quality, Software Quality Attributes and Specification, Cost of Quality, Defects, Faults, Failures, Defect Rate and Reliability, Defect Prevention, Reduction, and Containment, Overview of Different Types of Software Review, Introduction to Measurement and Inspection Process, Documents and Metrics.

UNIT-II:

Software Quality Metrics 8 Product Quality Metrics: Defect Density, Customer Problems Metric, Customer Satisfaction Metrics, Function Points, In-Process Quality Metrics: Defect Arrival Pattern, Phase-Based Defect Removal Pattern, Defect Removal Effectiveness, Metrics for Software Maintenance: Backlog Management Index, Fix Response Time, Fix Quality, Software Quality Indicators.

UNIT-III: Software Quality Management and Models 8 Modeling Process, Software Reliability Models: The Rayleigh Model, Exponential Distribution and Software Reliability Growth Models, Software Reliability Allocation Models, Criteria for Model Evaluation, Software Quality Assessment Models: Hierarchical Model of Software Quality Assessment.

UNIT-IV: Software Quality Assurance 12 Quality Planning and Control, Quality Improvement Process, Evolution of Software Quality Assurance (SQA), Major SQA Activities, Major SQA Issues, Zero Defect Software, SQA Techniques, Statistical Quality Assurance, Total Quality Management, Quality Standards and Processes.

UNIT-V: Software Verification, Validation & Testing: 8 Verification and Validation, Evolutionary Nature of Verification and Validation, Impracticality of Testing all Data and Paths, Proof of Correctness, Software Testing, Functional, Structural and Error Oriented Analysis & Testing, Static and Dynamic Testing Tools, Characteristics of Modern Testing Tools

Text Books

1. Jeff Tian, Software Quality Engineering (SQE), Wiley
2. Stephen H. Kan, Metrics and Models in Software Quality Engineering, Addison-Wesley
3. M. Cotterell, Software Project Management, Tata McGraw-Hill Publication.
4. Royce, Software Project Management, Pearson Education
5. Kieron Conway, Software Project Management, Dreamtech Press
6. S. A. Kelkar, Software Project Management, PHI Publication.

Rural Technology and Community Development

Unit 1: Data Analysis and Measures of Central Tendency- Meaning, nature, scope and limitations of statistics, collection of statistical data, classification, tabulation and diagrammatic representation of data, Measures of central tendency: Statistical averages Mean, Median, Mode.

Unit 2: Data, Information and Knowledge; concept of information, need of information (professional, educational, research), qualities of information, value of information, difference between data and information, properties of the needed information. Information and Management; planning, organizing, co-ordinating and controlling,

Unit 3: Concepts of marketing; difference between marketing selling and retailing; marketing mix, market-segmentation, marketing planning. Strategy and Approaches; modern concept of marketing.

Unit 4: Community development; concept, definition, meaning, need, history, principles, objectives and scope. Community Building: Coming of Age, Regenerating Community, Community Model.

Unit 5: Consensus Organizing Model, What's Behind Building Healthy Communities? Participatory Democracy, The Role of various NGOs in Community Development. The Role of Business and Government in Community Development Initiatives How to Form a Non-profit Corporation Fund Raising and Grant Writing.

Text/Reference Books:

1. Biddle, William Wishart. 1968. Encouraging Community Development: A Training Guide for Local Workers. New York: Holt, Rinehart and Winston.
2. Clark, Kenneth B. and Jeannette Hopkins, eds. 1969. A Relevant War Against Poverty: A Study of Community Action Programs and Observable Social Change. New York: Harper and Row.
3. Clinard, Marshall Barron. 1970. Slums and Community Development: Experiments in Self-Help
New York: Free Press.
4. Creevey, Lucy E., ed. 1986. Women Farmers in Africa: Rural Development in Mali and the Sahel. Syracuse, NY: Syracuse University Press.
5. Dobyns, Henry F., Paul L. Doughty, and Harold D. Lasswell, eds. 1971. Peasants, Power, and Applied Social Change: Vicos as a Model. Beverly Hills, CA: Sage.

7. Edwards, Allen David and Dorothy G. Jones. 1976. Community and Community Development. The Hague, Netherlands: Mouton.
8. Green, Tova and Peter Woodrow. 1994. Insight and Action: How to Discover and Support a Life of Integrity and Commitment to Change. Philadelphia, PA: New Society Publishers.
9. Heskin, Allen David. 1991. The Struggle for Community. Boulder, CO: Westview Press.
Kramer, Ralph M. and Harry Specht. 1975. Readings in Community Organization Practice. 2d ed. Englewood Cliffs, NJ: Prentice-Hall.
10. Lean, Mary. 1995. Bread, Bricks, and Belief: Communities in Charge of Their Future. West Hartford, CT: Kumarian Press.
11. Sustainable Rural Technology, by M.S. Viridi, Daya Publishing House, ISBN: 8170355656

Software Testing And Audit

Unit-I

Characterization of Distributed Systems: Introduction, Examples of distributed Systems, Resource sharing and the Web Challenges. Architectural models, Fundamental Models. Theoretical Foundation for Distributed System: Limitation of Distributed system, absence of global clock, shared memory, Logical clocks, Lamport's & vectors logical clocks. Concepts in Message Passing Systems: causal order, total order, total causal order, Techniques for Message Ordering, Causal ordering of messages, global state, termination detection

Unit-II

Distributed Mutual Exclusion: Classification of distributed mutual exclusion, requirement of mutual exclusion theorem, Token based and non token based algorithms, performance metric for distributed mutual exclusion algorithms. Distributed Deadlock Detection: system model, resource Vs communication deadlocks, deadlock prevention, avoidance, detection & resolution, centralized dead lock detection, distributed dead lock detection, path pushing algorithms, edge chasing algorithms.

Unit-III

Agreement Protocols: Introduction, System models, classification of Agreement Problem, Byzantine agreement problem, Consensus problem, Interactive consistency Problem, Solution to Byzantine Agreement problem, Application of Agreement problem, Atomic Commit in Distributed Database system. Distributed Resource Management: Issues in distributed File Systems, Mechanism for building distributed file systems, Design issues in Distributed Shared Memory, Algorithm for Implementation of Distributed Shared Memory.

Unit-IV

Failure Recovery in Distributed Systems: Concepts in Backward and Forward recovery, Recovery in Concurrent systems, Obtaining consistent Checkpoints, Recovery in Distributed Database Systems. Fault Tolerance: Issues in Fault Tolerance, Commit Protocols, Voting protocols, Dynamic voting protocols

Unit-V

Transactions and Concurrency Control: Transactions, Nested transactions, Locks, Optimistic Concurrency control, Timestamp ordering, Comparison of methods for concurrency control. Distributed Transactions: Flat and nested distributed transactions, Atomic Commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery. Replication: System model and group communication, Fault - tolerant services, highly available services, Transactions with replicated data

Referenece

Text books:

1. Singhal & Shivaratri, "Advanced Concept in Operating Systems", McGraw Hil
- 1 2. Ramakrishna,Gehrke," Database Management Systems", McGraw Hill
3. Vijay K.Garg Elements of Distributed Computing , Wiley
4. Coulouris, Dollimore, Kindberg, "Distributed System: Concepts and Design", Pearson Education
5. Tenanuanbaum, Steen," Distributed Systems", PHI

Digital Image Processing

UNIT-I

Introduction and Fundamentals

Motivation and Perspective, Applications, Components of Image Processing System, Element of Visual Perception, A Simple Image Model, Sampling and Quantization.

Image Enhancement in Spatial Domain

Introduction; Basic Gray Level Functions – Piecewise-Linear Transformation Functions: Contrast Stretching; Histogram Specification; Histogram Equalization; Local Enhancement; Enhancement using Arithmetic/Logic Operations – Image Subtraction, Image Averaging; Basics of Spatial Filtering; Smoothing - Mean filter, Ordered Statistic Filter; Sharpening – The Laplacian.

UNIT-2

Image Enhancement in Frequency Domain

Fourier Transform and the Frequency Domain, Basis of Filtering in Frequency Domain, Filters – Low-pass, High-pass; Correspondence Between Filtering in Spatial and Frequency Domain; Smoothing Frequency Domain Filters

– Gaussian Lowpass Filters; Sharpening Frequency Domain Filters – Gaussian Highpass Filters; Homomorphic Filtering.

Image Restoration

A Model of Restoration Process, Noise Models, Restoration in the presence of Noise only-Spatial Filtering – Mean Filters : Arithmetic Mean filter, Geometric Mean Filter, Order Statistic Filters – Median Filter, Max and Min filters ; Periodic Noise Reduction by Frequency Domain Filtering – Bandpass Filters ; Minimum Mean-square Error Restoration

UNIT-3

Color Image Processing

Color Fundamentals, Color Models, Converting Colors to different models, Color Transformation, Smoothing and Sharpening, Color Segmentation.

Morphological Image Processing

Introduction, Logic Operations involving Binary Images, Dilation and Erosion, Opening and Closing, Morphological Algorithms – Boundary Extraction, Region Filling, Extraction of Connected Components, Convex Hull, Thinning, Thickening

UNIT-4

Registration

Introduction, Geometric Transformation – Plane-to-Plane transformation, Mapping, Stereo Imaging – Algorithms to Establish Correspondence, Algorithms to Recover Depth

Segmentation

Introduction, Region Extraction, Pixel-Based Approach, Multi-level Thresholding, Local Thresholding, Region-based Approach. Edge and Line Detection : Edge Detection, Edge Operators, Pattern Fitting Approach, Edge Linking and Edge Following, Edge Elements Extraction by Thresholding, Edge Detector Performance, Line Detection, Corner Detection.

UNIT-5

Feature Extraction

Representation, Topological Attributes, Geometric Attributes

Description

Boundary-based Description, Region-based Description, Relationship.

Object Recognition

Deterministic Methods, Clustering, Statistical Classification, Syntactic Recognition, Tree Search, Graph Matching

Text Books

- 1) Digital Image Processing 2nd Edition, Rafael C. Gonzalvez and Richard E. Woods. Published by: Pearson Education
- 2) Digital Image Processing and Computer Vision, R.J. Schalkoff. Published by: John Wiley and Sons, NY.
- 3) Fundamentals of Digital Image Processing, A.K. Jain. Published by Prentice Hall, Upper Saddle River, NJ.

DISTRIBUTED SYSTEMS

Unit –I

Characterization of Distributed Systems: Introduction, Examples of distributed systems, Resource sharing and the Web Challenges.

System Models: Architectural models, Fundamental Models

Theoretical Foundation for Distributed System: Limitation of Distributed system, absence of global clock, shared memory, Logical clocks, Lamport's & vectors logical clocks, Causal ordering of messages, global state, termination detection.

Distributed Mutual Exclusion: Classification of distributed mutual exclusion, requirement of mutual exclusion theorem, Token based and non token based algorithms, performance metric for distributed mutual exclusion algorithms.

Unit –II

Distributed Deadlock Detection: system model, resource Vs communication deadlocks, deadlock prevention, avoidance, detection & resolution, centralized dead lock detection, distributed dead lock detection, path pushing algorithms, edge chasing algorithms.

Agreement Protocols: Introduction, System models, classification of Agreement Problem, Byzantine agreement problem, Consensus problem, Interactive consistency Problem, Solution to Byzantine Agreement problem, Application of Agreement problem, Atomic Commit in Distributed Database system.

Unit –III

Distributed Objects and Remote Invocation: Communication between distributed objects, Remote procedure call, Events and notifications, Java RMI case study.

Security: Overview of security techniques, Cryptographic algorithms, Digital signatures

Cryptography pragmatics, Case studies: Needham Schroeder, Kerberos, SSL & Millicent.

Distributed File Systems: File service architecture, Sun Network File System, The Andrew File System, Recent advances.

Unit –IV

Transactions and Concurrency Control: Transactions, Nested transactions, Locks, Optimistic Concurrency control, Timestamp ordering, Comparison of methods for concurrency control. **Distributed Transactions:** Flat and nested distributed transactions, Atomic Commit protocols, Concurrency control in distributed transactions, Distributed deadlocks, Transaction recovery. Replication: System model and group communication, Fault -tolerant services, highly available services, Transactions with replicated data.

Unit –V

Distributed Algorithms: Introduction to communication protocols, Balanced sliding window protocol, Routing algorithms, Destination based routing, APP problem, Deadlock free Packet switching, Introduction to Wave & traversal algorithms, Election algorithm. **CORBA Case Study:** CORBA RMI, CORBA services.

Books:

1. Singhal & Shivaratri, "Advanced Concept in Operating Systems", McGraw Hill
2. Coulouris, Doll more, Kindberg, "Distributed System: Concepts and Design", Pearson Ed.
3. Gerald Tel, "Distributed Algorithms", Cambridge University Press

REAL TIME SYSTEMS

UNIT-I: Introduction

Definition, Typical Real Time Applications: Digital Control, High Level Controls, Signal Processing etc., Release Times, Deadlines, and Timing Constraints, Hard Real Time Systems and Soft Real Time Systems, Reference Models for Real Time Systems: Processors and Resources, Temporal Parameters of Real Time Workload, Periodic Task Model, Precedence Constraints and Data Dependency.

UNIT-II: Real Time Scheduling

Common Approaches to Real Time Scheduling: Clock Driven Approach, Weighted Round Robin Approach, Priority Driven Approach, Dynamic Versus Static Systems, Optimality of Effective-Deadline-First (EDF) and Least-Slack-Time-First (LST) Algorithms, Offline Versus Online Scheduling, Scheduling Aperiodic and Sporadic jobs in Priority Driven and Clock Driven Systems.

UNIT-III: Resources Access Control

Effect of Resource Contention and Resource Access Control (RAC), Non-preemptive Critical Sections, Basic Priority-Inheritance and Priority-Ceiling Protocols, Stack Based Priority-Ceiling Protocol, Use of Priority-Ceiling Protocol in Dynamic Priority Systems, Preemption Ceiling Protocol, Access Control in Multiple-Unit Resources, Controlling Concurrent Accesses to Data Objects.

UNIT-IV: Multiprocessor System Environment

Multiprocessor and Distributed System Model, Multiprocessor Priority-Ceiling Protocol, Schedulability of Fixed-Priority End-to-End Periodic Tasks, Scheduling Algorithms for End-to-End Periodic Tasks, End-to-End Tasks in Heterogeneous Systems, Predictability and Validation of Dynamic Multiprocessor Systems, Scheduling of Tasks with Temporal Distance Constraints.

UNIT-V: Real Time Communication

Model of Real Time Communication, Priority-Based Service and Weighted Round-Robin Service Disciplines for Switched Networks, Medium Access Control Protocols for Broadcast Networks, Internet and Resource Reservation Protocols, Real Time Protocols, Communication in Multicomputer System, An Overview of Real Time Operating Systems.

Books:

- 1) Real Time Systems by Jane W. S. Liu, Pearson Education Publication.
- 2) Real-Time Systems: Scheduling, Analysis, and Verification by Prof. Albert M. K. Cheng, John Wiley and Sons Publications.

PATTERN RECOGNITION

Unit-I

Introduction:

Basics of pattern recognition, Design principles of pattern recognition system, Learning and adaptation, Pattern recognition approaches, Mathematical foundations – Linear algebra, Probability Theory, Expectation, mean and covariance, Normal distribution, multivariate normal densities, Chi squared test.

Unit-II

Statistical Pattern Recognition:

Bayesian Decision Theory, Classifiers, Normal density and discriminant functions,

Unit – III

Parameter estimation methods: 12 Maximum-Likelihood estimation, Bayesian Parameter estimation, Dimension reduction methods - Principal Component Analysis (PCA), Fisher Linear discriminant analysis, Expectation-maximization (EM), Hidden Markov Models (HMM), Gaussian mixture models.

Unit - IV

Nonparametric Techniques:

Density Estimation, Parzen Windows, K-Nearest Neighbor Estimation, Nearest Neighbor Rule, Fuzzy classification.

Unit - V

Unsupervised Learning & Clustering:

Criterion functions for clustering, Clustering Techniques: Iterative square - error partitional clustering – K means, agglomerative hierarchical clustering, Cluster validation.

REFERENCES:

1. Richard O. Duda, Peter E. Hart and David G. Stork, "Pattern Classification", 2nd Edition, John Wiley, 2006.
2. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2009.
3. S. Theodoridis and K. Koutroumbas, "Pattern Recognition", 4th Edition, Academic Press, 2009.

CLUSTER COMPUTING

UNIT I:

Basic concepts in Distributed Systems

Notion of time Distributed Mutual exclusion, Consensus, Failure models Paradigms for process interaction in distributed programs, Programming Paradigms, Shared memory, Message passing, Workflows

UNIT II:

Introduction to Cluster Computing, Cluster Middleware: An Introduction, Early Cluster Architecture and High Throughput Computing Clusters, Networking, Protocols and I/O for Clusters, Setting Up and Administering a Cluster

UNIT III:

Cluster Technology for High Availability, Performance Models and Simulation, Process Scheduling, Load Sharing and Load Balancing, Distributed Shared Memory,

UNIT IV:

Introduction to Grid Architecture, Characterization of Grid, and Grid related standard bodies, Grid types, Topologies, Components and Layers, Comparison with other approaches.

UNIT V:

System Infrastructure, Traditional paradigms for distributed computing, Web Services, Grid standards:

OGSA and WSRF, Case Studies of Cluster Systems: Beowulf, COMPaS, NanOS and PARAM

REFERENCES:

1. Grid and Cluster Computing, Prabhu C.S.R, PHI Learning Private Limited
2. A networking Approach To Grid Computing by Daniel Minoli (Chapter 1) (John Wiley and Sons, INC Publication)
3. Distributed and Cloud Computing, First Edition, Geoffrey C. Fox, KaiHwang, Jack J. Dongarra, Elsevier India Pvt. Ltd.-New Delhi
4. Fran Berman, Geoffrey C. Fox, Anthony J.G Hey Grid Computing making the global infrastructure a Reality

5. High Performance Cluster Computing: Architectures and Systems, Vol. 1, Prentice Hall
6. In search of clusters (2nd ed.), Gregory F. Pfister, IBM, Austin, TX, Prentice-Hall

GRID COMPUTING

UNIT I

CONCEPTS AND ARCHITECTURE¹⁰

Introduction-Parallel and Distributed Computing-Cluster Computing-Grid Computing- Anatomy and Physiology of Grid-Review of Web Services-OGSA-WSRF.

UNIT II

GRID MONITORING

Grid Monitoring Architecture (GMA) - An Overview of Grid Monitoring Systems- GridICE – JAMM - MDS-Network Weather Service-R-GMA-Other Monitoring Systems- Ganglia and GridMon

UNIT III

GRID SECURITY AND RESOURCE MANAGEMENT

Grid Security-A Brief Security Primer-PKI-X509 Certificates-Grid Security-Grid Scheduling and Resource Management-Scheduling Paradigms- Working principles of Scheduling -A Review of Condor, SGE, PBS and LSF-Grid Scheduling with QoS.

UNIT IV

DATA MANAGEMENT AND GRID PORTALS¹⁰

Data Management-Categories and Origins of Structured Data-Data Management Challenges-Architectural Approaches-Collective, Data Management Services-Federation Services-Grid Portals-First-Generation Grid Portals-Second-Generation Grid Portals.

UNIT V

GRID MIDDLEWARE

List of globally available Middlewares - Case Studies-Recent version of Globus Toolkit and gLite - Architecture, Components and Features

REFERENCES:

3. JoshyJoseph, CraigFellenstein—Grid Computing, Pearson Education, 2004.
4. Vladimir Silva—Grid Computing for Developers,DreamtechPress, 2006.
3. Fran Berman, Geoffrey C. Fox, Anthony J.G Hey Grid Computing making the global infrastructure a Reality, Wiley
3. AhmarAbbas--Grid Computing —A Practical Guide to Technology and Applications, Firewall Media, 2006.

ENTREPRENEURSHIP DEVELOPMENT

UNIT -I

Entrepreneurship- definition. growth of small scale industries in developing countries and their positions vis-a -vis large industries; role of small scale industries in the national economy; characteristics and types of small scale industries; demand based and resources based ancillaries and sub-control types. Government policy for small scale industry; stages in starting a small scale industry.

UNIT -II

Project identification- assessment of viability, formulation, evaluation, financing, field-study and collection of information, preparation of project report, demand analysis, material balance and output methods, benefit cost analysis, discounted cash flow, internal rate of return and net present value methods. 8

UNIT -III Accountancy- Preparation of balance sheets and assessment of economic viability, decision making, expected costs, planning and production control, quality control, marketing, industrial relations, sales and purchases, advertisement, wages and incentive, inventory control, preparation

of financial reports, accounts and stores studies.

UNIT -IVProject Planning and control: The financial functions, cost of capital approach in project planning and control. Economic evaluation, risk analysis, capital expenditures, policies and practices in public enterprises. profit planning and programming, planning cash flow, capital expenditure and operations. control of financial flows, control and communication.

U N I T -V

Laws concerning entrepreneur viz, partnership laws, business ownership, sales and income taxes and workman compensation act. Role of various national and state agencies which render assistance to small scale industries. 2

Text / Reference Books:

1. Forbat, John, "Entrepreneurship" New Age International.
2. Havinal, Veerbhadrappa, "Management and Entrepreneurship" New Age International
3. Joseph, L. Massod, "Essential of Management", Prentice Hall of India.

Introduction to Bio Technology

UNIT-I Introduction: Concept nature and scope of biotechnology. Cell Structure and Function: Eukaryotic and prokaryotic cells, cell wall, membrane organization, cell organelles, Nucleus, Mitochondria, endoplasmic reticulum, chloroplast, viruses and toxins into cells. Cell Division: Mitosis and Meiosis.

UNIT-II

Biomolecules: A brief account of structure of carbohydrates, Lipids and Proteins. Genes: Brief idea about Mendel's laws and chromosomes, nature of genetic materials, DN A and RNA, DNA replication.

UNIT-III

Gene Expression: Central dogma, genetic code, molecular mechanism on mutations, regulations of gene expression, house keeping genes, differentiation and development mutations and their molecular basic. Genetic Engineering: Introduction, cloning (vectors and enzymes), DNA and genomic libraries, Transgenics, DNA fingerprinting, genomics.

UNIT-IV

Applications of Biotechnology: Bioprocess and fermentation technology, cell culture, Enzyme technology, biological fuel generation, sewage treatment, environmental biotechnology, biotechnology and medicine, biotechnology in agriculture, food and beverage technology, production of biological invention.

UNIT-V

Safety and Ethics: Safety, social, moral and ethic considerations, environmental ethics, bioethics and stem cell research, safety of new biotechnology foods, agro biodiversity and donor policies. Text Books/

Reference Books:

1. Smith, "Biotechnology" Cambridge Press.
2. P.K. Gupta, "Elements of Biotechnology" Rastogi
3. H. D. Kumar, "Modern concepts of Biotechnology" Vikas publishing House

ETHICAL HACKING AND PREVENTION

Unit-I

Ethical Hacking: Introduction, Networking & Basics, Foot Printing, Google Hacking, Scanning, Windows Hacking, Linux Hacking, Trojans & Backdoors, Virus & Worms,

Unit-II

Proxy & Packet Filtering, Denial of Service, Sniffer, Social Engineering System and Network Vulnerability and Threats to Security , Various types of attack and the various types of attackers in the context of the vulnerabilities associated with computer and information systems and networks Physical Security, Steganography,

Unit-III

Cryptography, Wireless Hacking, Firewall & Honeypots, IDS & IPS, Vulnerability, Penetration Testing, Session Hijacking, Hacking Web Servers, SQL Injection, Cross Site Scripting, Exploit Writing, Buffer Overflow,

Unit-IV

Reverse Engineering, Email Hacking, Incident Handling & Response, Bluetooth Hacking, Mobile Phone Hacking Basic ethical hacking tools and usage of these tools in a professional environment. Legal, professional and ethical issues likely to face the domain of ethical hacking. Ethical responsibilities, professional integrity and making appropriate use of the tools and techniques associated with ethical hacking.

REFERENCES:

1. Dominic Chell , Tyrone Erasmus, Shaun Colley, Ofllie Whitehouse, The Mobile Application Hacker's Handbook , Wiley
2. Michael Gregg, "Certified Ethical Hacker (CEH) Cert Guide", Pearson India, 2014
3. Rafay Baloch, "Ethical Hacking and Penetration Testing Guide" CRC Press
4. Allen Harper , Shome Harris, Jonathan Ness ,Chris Eagle, Gideon Lenkey,TerronVilliams "Gray Hat Hacking The Ethical Hakers Handbook." TMH
5. Patrick Engebretson, "The Basics of Hacking and Penetration Testing, Second Edition:Ethical Hacking and Penetration Testing Made Easy, 2nd Edition, Elsevier
6. Jon Erickson "HACKING, The art of Exploitation", William Pollock.

NON-CONVENTIONAL ENERGY RESOURCES

UNIT-I Introduction Various non-conventional energy resources- Introduction, availability, classification, relative merits and demerits. Solar Cells:Theory of solar cells. solar cell materials, solar cell array, solar cell power plant, limitations.

UNIT-II

Solar Thermal Energy: Solar radiation, flat plate collectors and their materials, applications and performance, focussing of collectors and their materials, applications and performance; solar thermal power plants, thermal energy storage for solar heating and cooling, limitations.

UNIT-III Geothermal Energy: Resources of geothermal energy, thermodynamics of geo-thermal energy conversion-electrical conversion, non-electrical conversion, environmental considerations. 4 Magneto-hydrodynamics (MHD): Principle of working of MHD Power plant, performance and limitations. Fuel Cells: Principle of working of various types of fuel cells and their working, performance and limitations.

UNIT-IV Thermo-electrical and thermionic Conversions: Principle of working, performance and limitations. Wind Energy: Wind power and its sources, site selection, criterion, momentum theory, classification of rotors, concentrations and augments, wind characteristics. performance and limitations of energy conversion systems.

UNIT-V

Bio-mass:Availability of bio-mass and its conversion theory. Ocean Thermal Energy Conversion (OTEC):Availability, theory and working principle, performance and limitations.

Wave and Tidal Wave:Principle of working, performance and limitations.Waste Recycling Plants.

Text/References Books:

1. Raja etal, "Introduction to Non-Conventional Energy Resources" Scitech Publications.
2. John Twideu and Tony Weir, "Renewal Energy Resources" BSP Publications, 2006.
3. M.V.R. Koteswara Rao, " Energy Resources: Conventional & Non-Conventional " BSP Publications,2006.
4. D.S. Chauhan,"Non-conventional Energy Resources" New Age International. 5. C.S. Solanki, "Renewal Energy Technologies: A Practical Guide for Beginners" PHI

Learning.

8. Peter Auer, "Advances in Energy System and Technology". Vol. 1 & II Edited by Academic Press.

Digital Image Processing Lab

1. Implement the spatial image enhancement function on a bit map image-
 - a) Mirroring (Inversion)
 - b) Rotation (Clockwise)
 - c) Enlargement (Double size)
2. Implement
 - a) Low Pass Filter
 - b) High Pass Filter
3. Implement
 - a) Arithmetic Mean Filter
 - b) Geometric Mean Filter
4. Implement Smoothing and sharpening of an 8-bit color image.
5. Implement
 - a) Boundary Extraction Algorithm
 - b) Graham's Scan Algorithm.
6. Implement
 - a) Edge Detection
 - b) Line Detection.